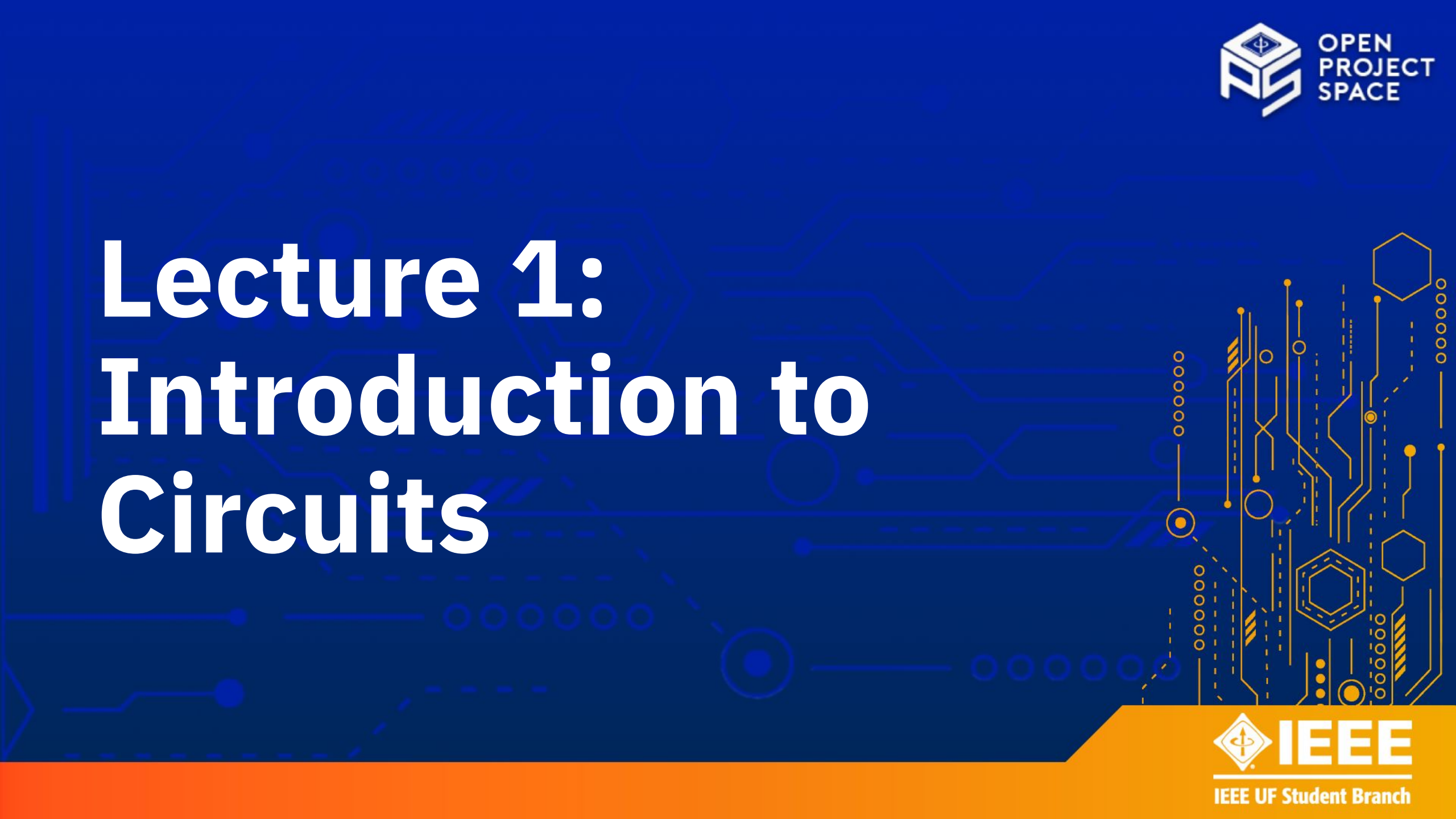
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Lecture 1: Introduction to Circuits



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Welcome to OPS!

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Today we will discuss:

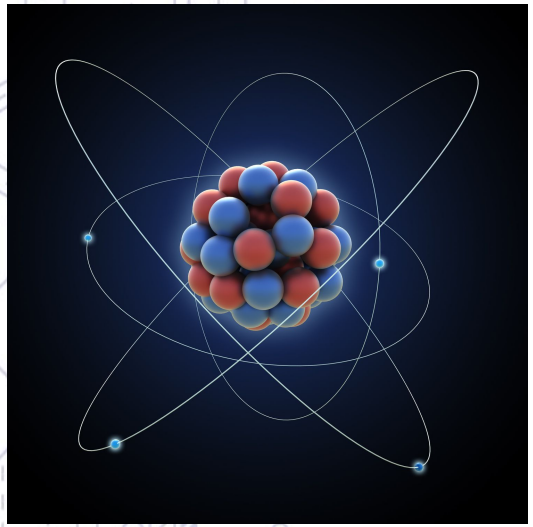
- Coordinator Introductions
- Electrons, charge, voltage, and current
- Conductivity and resistance
- Ohm’s Law
- LEDs, buttons, breadboards
- Circuit schematics
- Debugging
- OPS/IEEE Information

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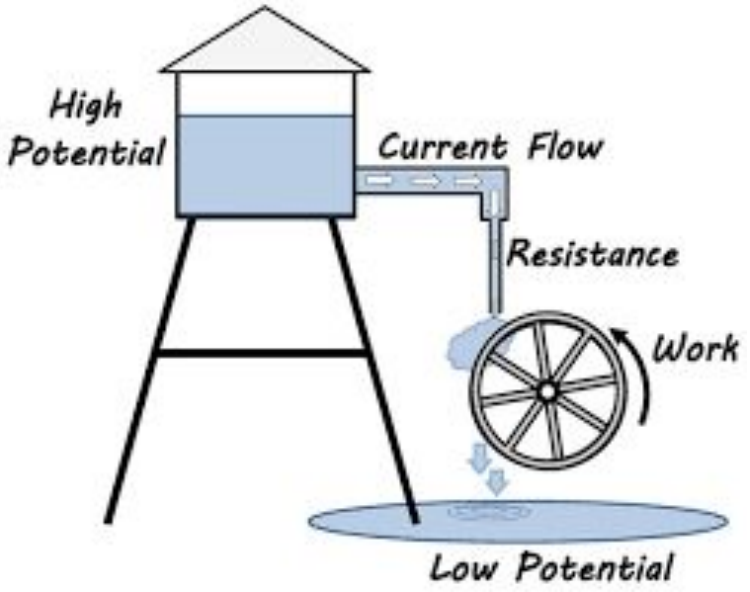
Fundamentals of Electricity

Atoms



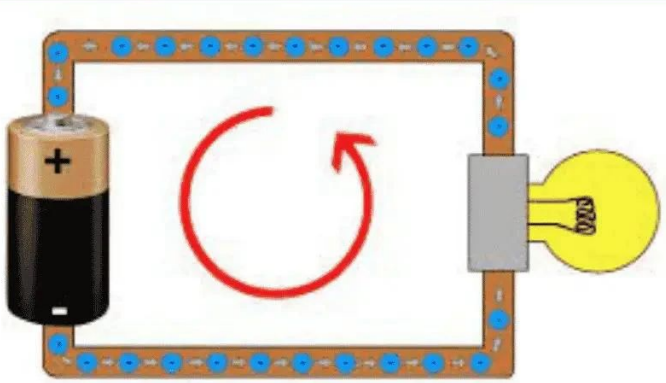
- **Atoms are made up of charged subatomic particles**
  - Protons
  - Electrons
- **1 Coulomb = charge of 6.24×10^{18} electrons**

Movement of Electrons



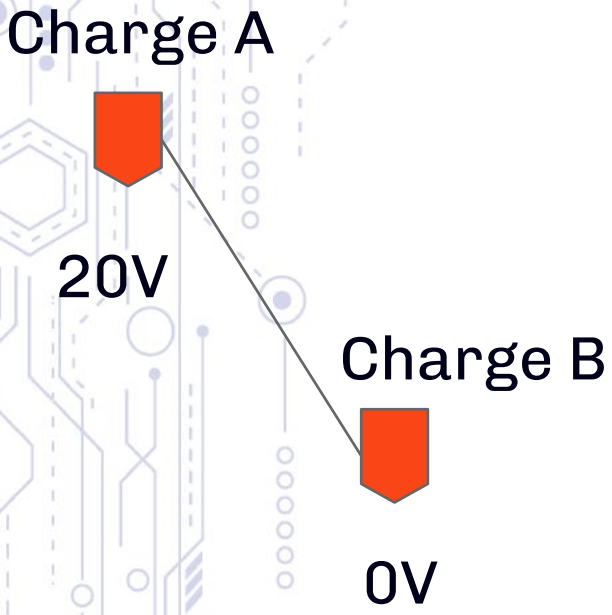
- **Electrons flow from a relative HIGH potential to a relative LOW potential**
- **Electrons flow better in certain materials and worse in others**
  - Conductors = better
  - Resistors = worse

Current (Amps)



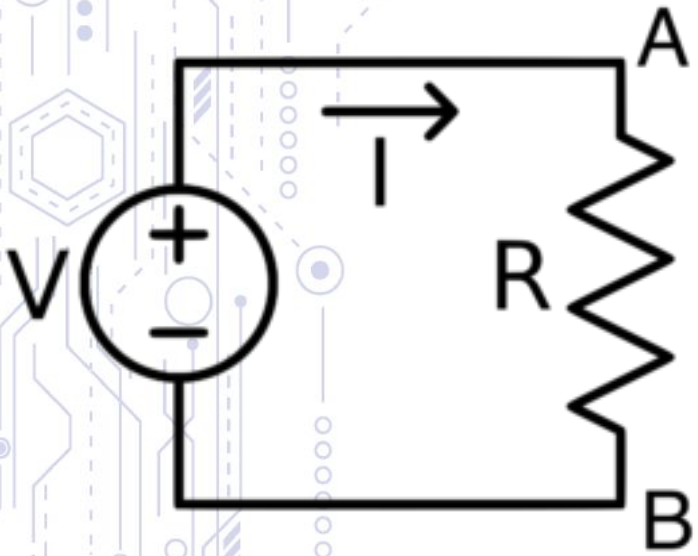
- **Recall that charge is measured in Coulombs**
- **Amperes (more commonly shorted to Amps) measure current**
  - 1 Amp = 1 Coulomb per second

Voltage (Volts)



- Voltage between two points describes how much ‘effort’ is needed for some charge to move from one point to another
- Example: 20V are needed to move charge A to charge B

Voltage (Volts)



- Voltage is the electrical potential difference between two points
- Unlike Coulombs (charge) or Amps (Coulombs per second), voltage is a relative measurement

Resistance (Ohms)



- Electrons in ANY medium collide with the atoms of the medium and lose ‘speed’
- This loss of speed can be measured as a <u>decrease</u> in current

Resistors and Heat





- **When electrons hit atoms via resistance, it causes the atoms to *jiggle***
- **Atoms ‘vibrating’ creates heat**
  - Good use case: electric heater
  - Bad use case: light bulb overheats

Units of Measurement

- Voltage is measured in Volts (V)
- Current is measured in Amps (A)
- Resistance is measured in Ohms (Ω)

Ohm’s Law

Ohm's Law



- **Named after Georg Ohm**
- **States that:**
  - Voltage = Current * Resistance
  - V = IR
- **Voltage (V) and Current (I) are directly proportional**
- **Voltage (V) and Resistance (R) are directly proportional**
- **Current (I) and Resistance (R) are inversely proportional**

Ohm's Law Example

$V = I * R$

$5V = 2A * R$

$2.5\ V/A = R$

$R = 2.5\ \Omega$

- **If we know two variables, we can solve for the third variable**
- **Example: The voltage across a wire reads 5V and the current through the wire reads 2A. What is the resistance through the wire?**

Check Your Understanding

Question 1/1

In your circuit, you measure a voltage drop of 10V across a 100Ω resistor. How much current is flowing through the resistor?

Check Your Understanding

Question 1/1

$V = I * R$

$10V = I * 100\Omega$

$0.1\ V/\Omega = I$

$I = 0.1\ A$

In your circuit, you measure a voltage drop of 10V across a 100Ω resistor. How much current is flowing through the resistor?

Ohm's Law: An OPS Example

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$V = I * R$

$5V = I * 220\ \Omega$

$0.0227\ A = I$

$I = 22.7\ mA$

$V = I * R$

$5V = I * 50\ \Omega$

$0.1\ A = I$

$I = 0.1\ A$

- **Imagine we light up an LED using a 5V power supply and a 220Ω resistor**
- **This results in a current of 0.0227A, or 22.7 mA, which is good for an LED!**
- **If we had only used a 50Ω resistor instead, the LED would burn out with 0.1A passing through it**

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Circuit Components

Circuit Components

The circuit components we will be covering in this lecture are:

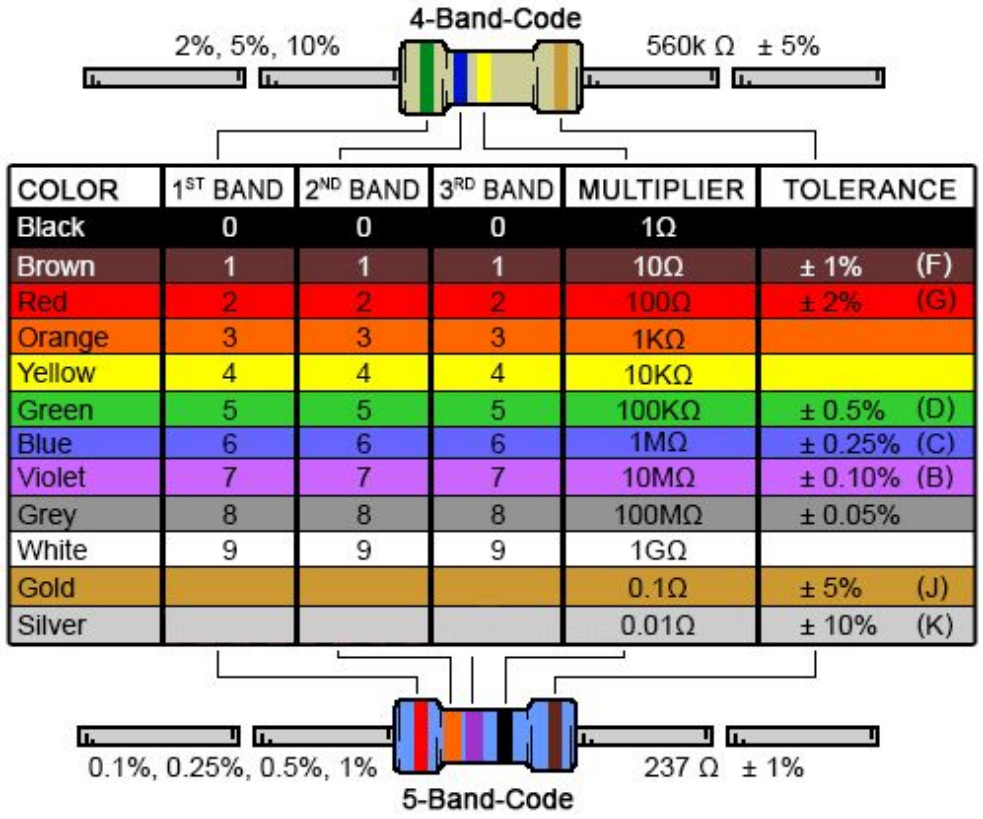
- Resistors
- LEDs
- Buttons
- Breadboards

Resistors



- Electrons in ANY medium collide with the atoms of the medium and lose ‘speed’
- This loss of speed can be measured as a decrease in current
- To control how much we decrease the current, we use resistors!
- Resistors come in different shapes, sizes, and values

Resistors



COLOR	1ST BAND	2ND BAND	3RD BAND	MULTIPLIER	TOLERANCE	
Black	0	0	0	1Ω		
Brown	1	1	1	10Ω	± 1%	(F)
Red	2	2	2	100Ω	± 2%	(G)
Orange	3	3	3	1KΩ		
Yellow	4	4	4	10KΩ		
Green	5	5	5	100KΩ	± 0.5%	(D)
Blue	6	6	6	1MΩ	± 0.25%	(C)
Violet	7	7	7	10MΩ	± 0.10%	(B)
Grey	8	8	8	100MΩ	± 0.05%	
White	9	9	9	1GΩ		
Gold				0.1Ω	± 5%	(J)
Silver				0.01Ω	± 10%	(K)

- Resistors are color coded
- Each color corresponds to a value and a multiplier
- The Digikey Resistor Color Code Calculator can be used to find the value of a resistor based on its colors!

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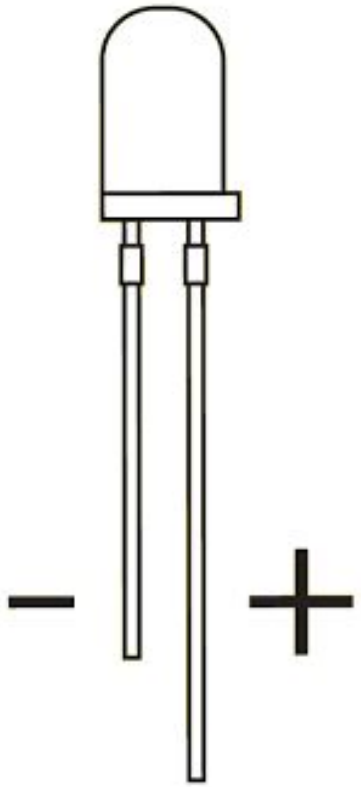
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LEDs



- Diodes have two terminals and allows the flow of current in only one direction
- Because of this, we classify diodes as polarized components
- Light Emitting Diodes (LEDs) fall under this category as well
- Passing sufficient current through the LED produces light

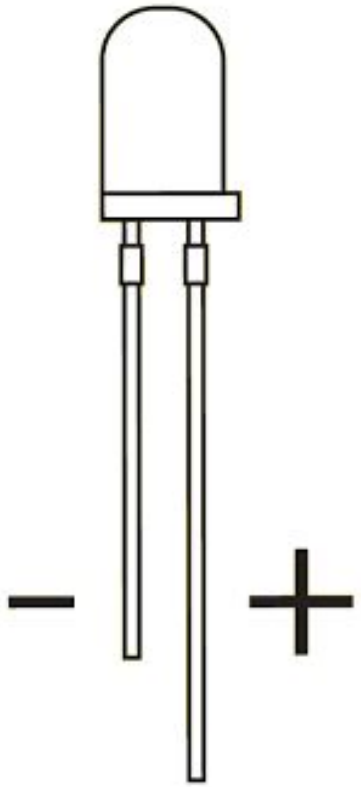
LEDs



- **LEDs have two legs:**
  - **Anode (+)** is the **long end**
  - **Cathode (-)** is the **short end**
- **The anode is connected to the current source, and the cathode is connected to ground**
- **Remember: “Cats like the ground”**

LEDs

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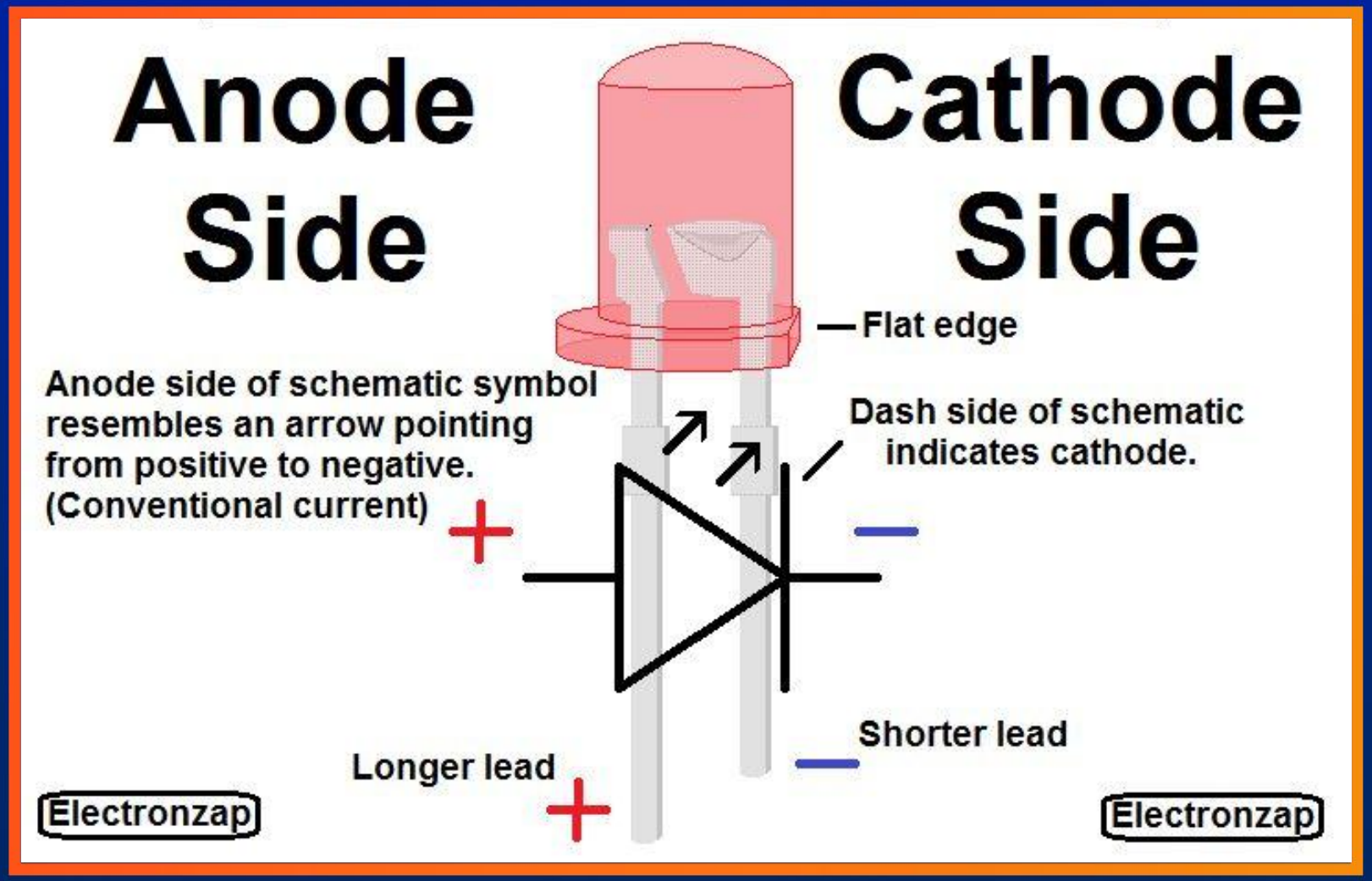


LED orientation is one of two leading issues in OPS when it comes to circuits not working

ALWAYS REMEMBER TO CHECK YOUR LED ORIENTATION!!!

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Credit: Electronzap Pinterest

https://i.pinimg.com/736x/45/07/85/450785a6db44e22ec3af85599e91535c.jpg

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Buttons



- When a button is pressed, a circuit is closed with a conductive material
- The wire is connected via the pressable part of the button

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Button Diagram

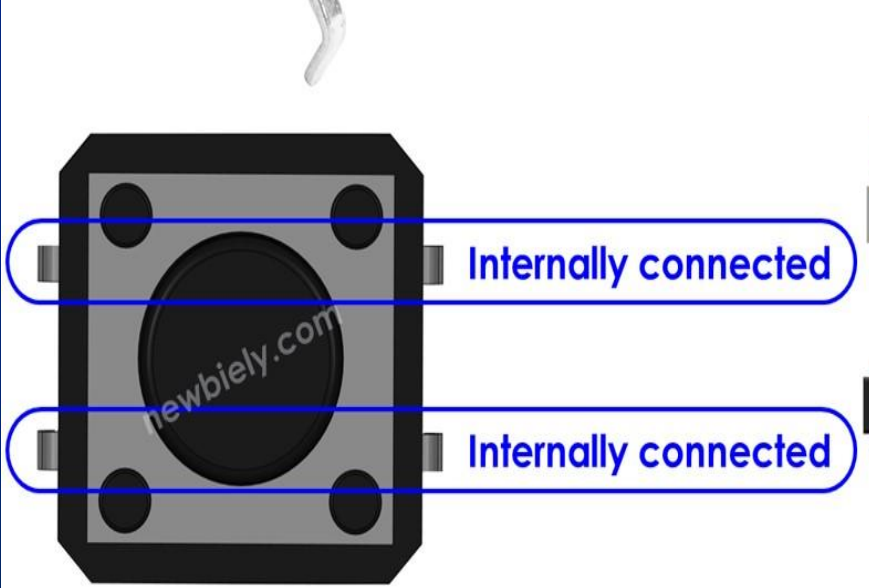


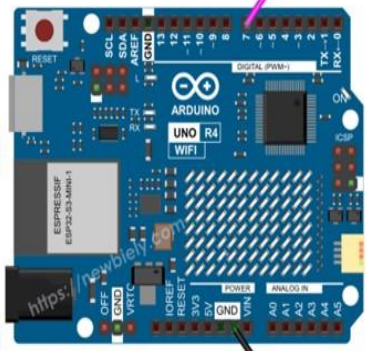


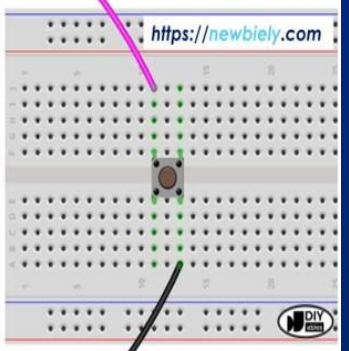










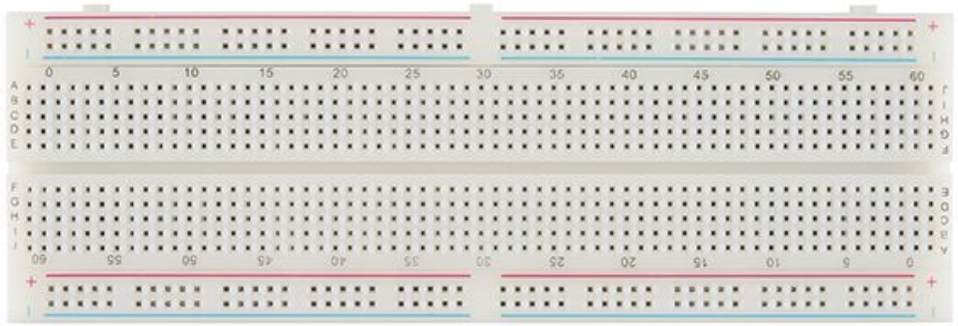


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Breadboards

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- Breadboards allow you to build circuits in a flexible manner
- Rewiring is easy; no soldering, no PCB

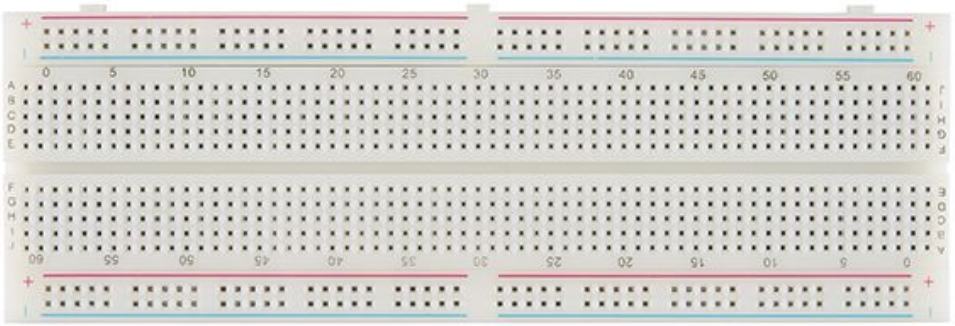


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Breadboards

- **Follow the rules of the board!**



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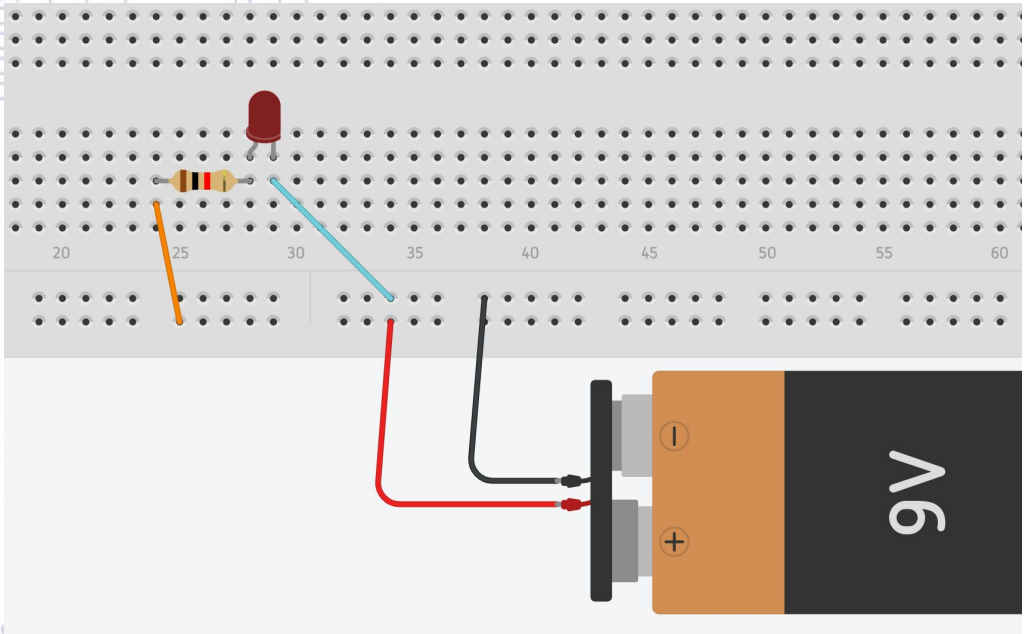
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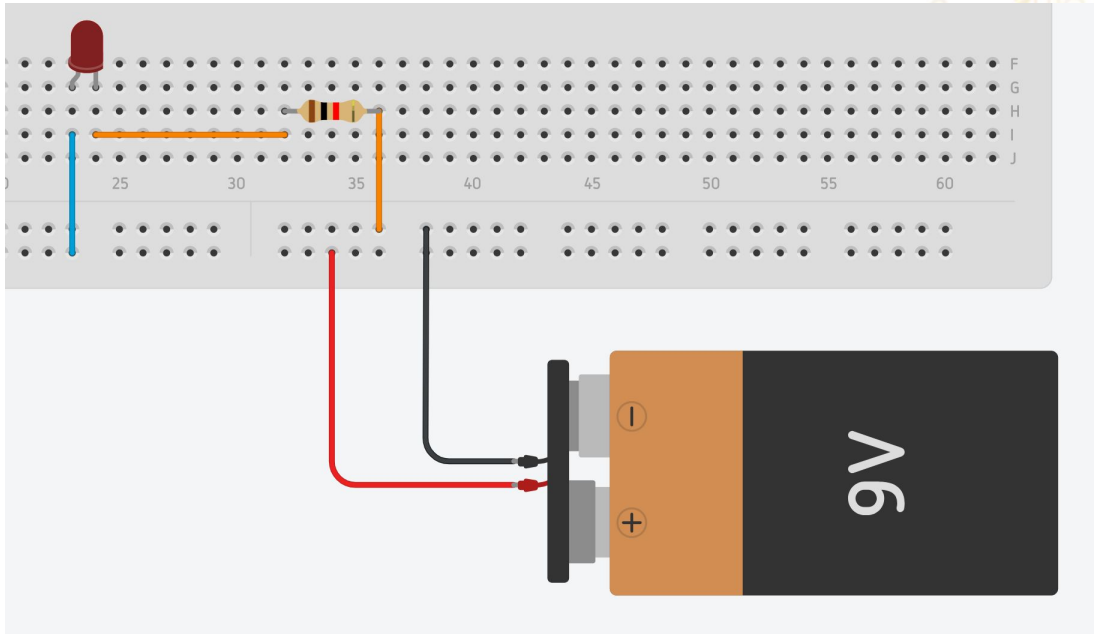
Check Your Understanding

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Question 1/2

Which is the correct resistor placement?





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Check Your Understanding

Question 2/2

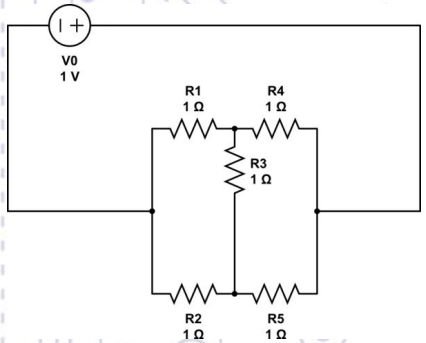


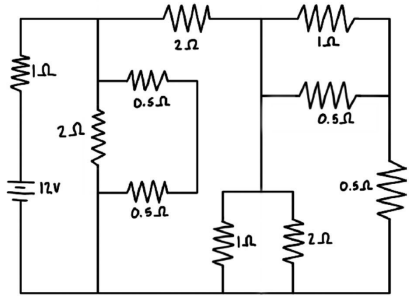
What went wrong here?

https://www.instagram.com/p/C_5txqIRMgC/

Schematics

Schematics

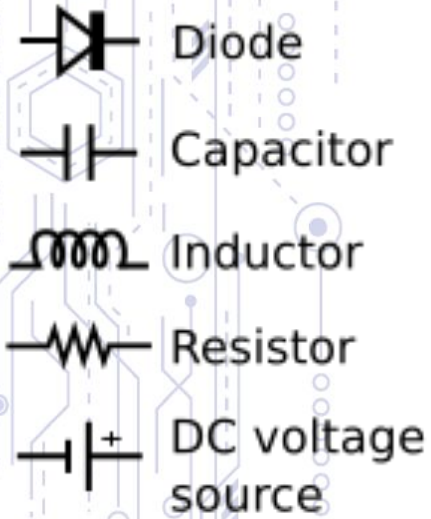


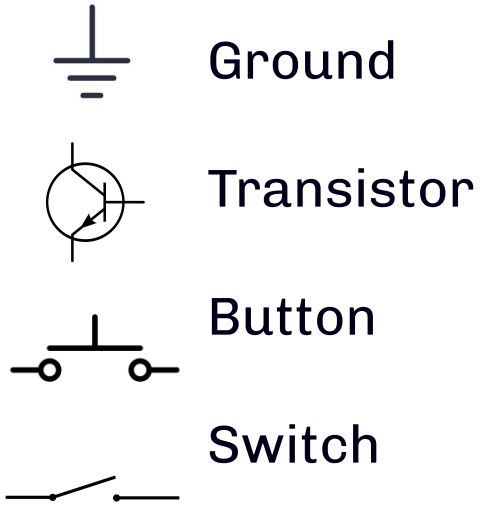


- **Schematics are organized circuit diagrams**
  - You will see these all the time!
- **Schematics use symbols to represent circuit components**

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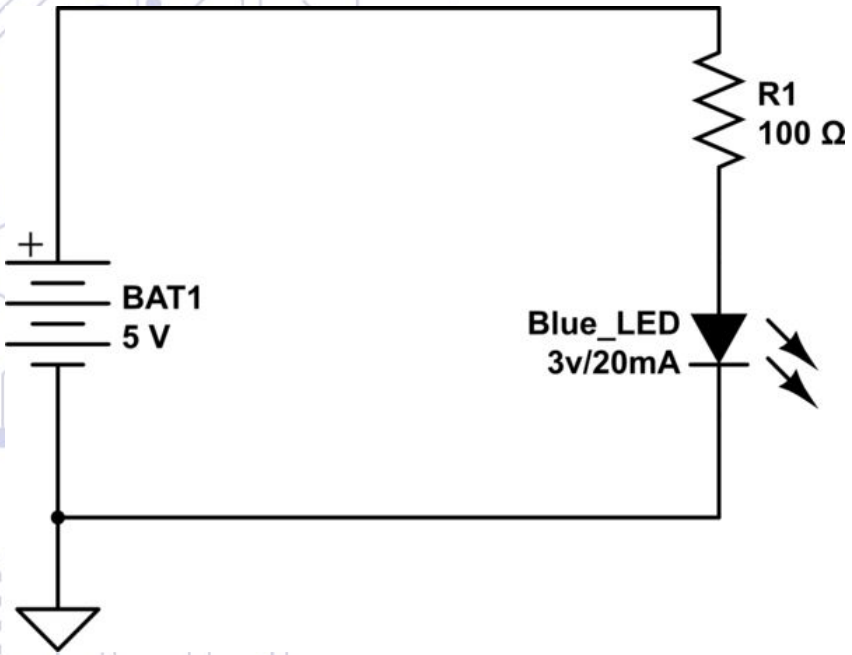
Circuit Symbols





- These are some common symbols and the components that they represent
- Get familiar with these symbols!
- Note: Symbols may vary based on style, component type, etc

Reading Schematics



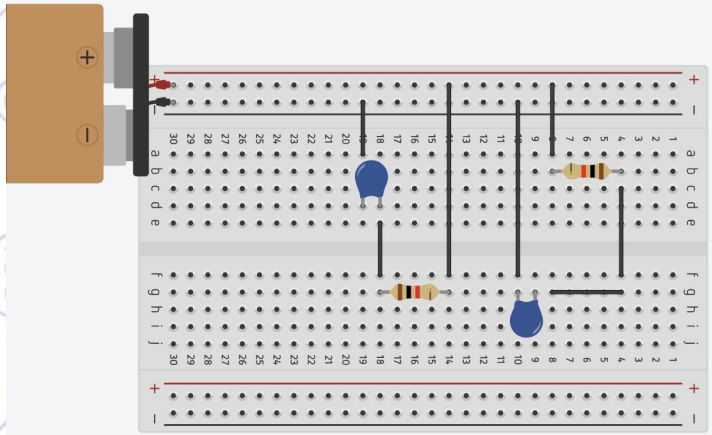
- **This is a simple LED circuit schematic**
- **Notice the numerical details and variations to symbols**
- **A good schematic is:**
  - Easy to understand
  - Easy to replicate
  - Easy to fabricate

Check Your Understanding

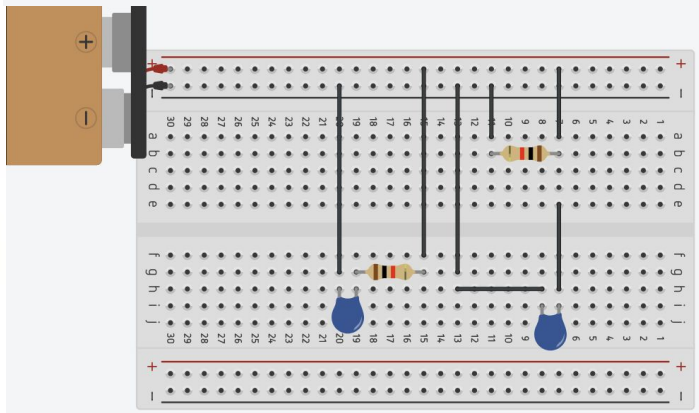
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Question 1/1

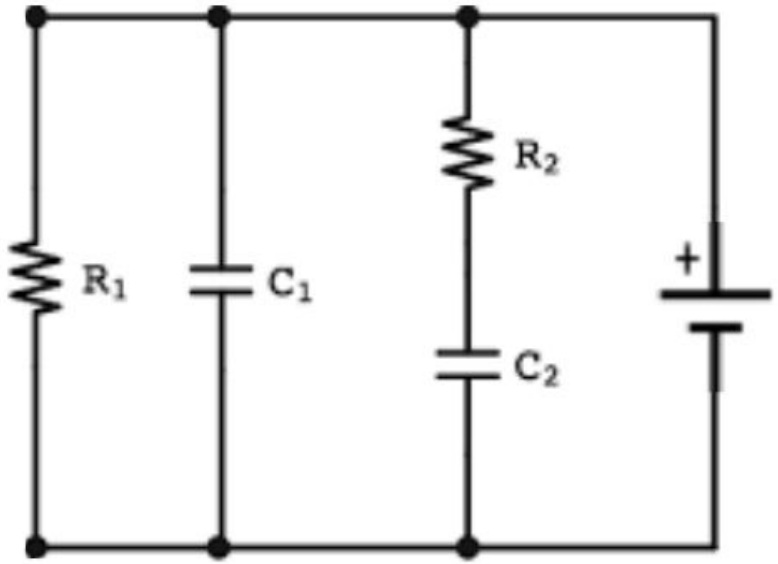
A



B



Which breadboard correctly follows the schematic?



Correct Answer:

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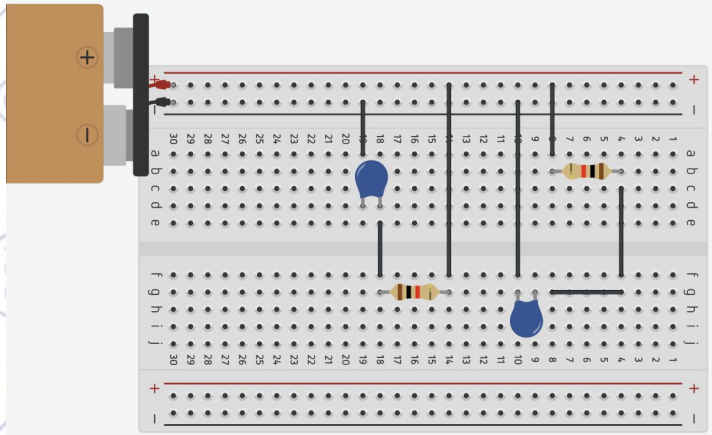
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Check Your Understanding

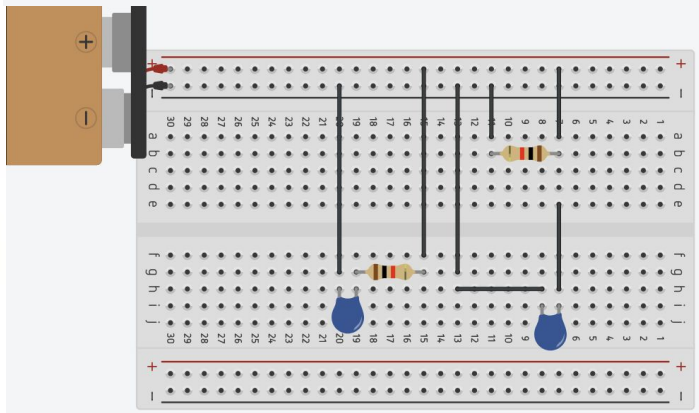
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Question 1/1

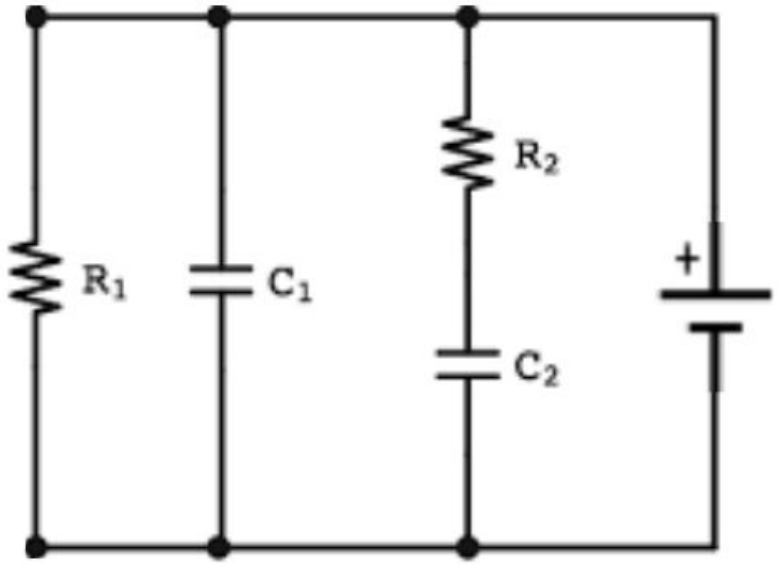
A



B



Which breadboard correctly follows the schematic?



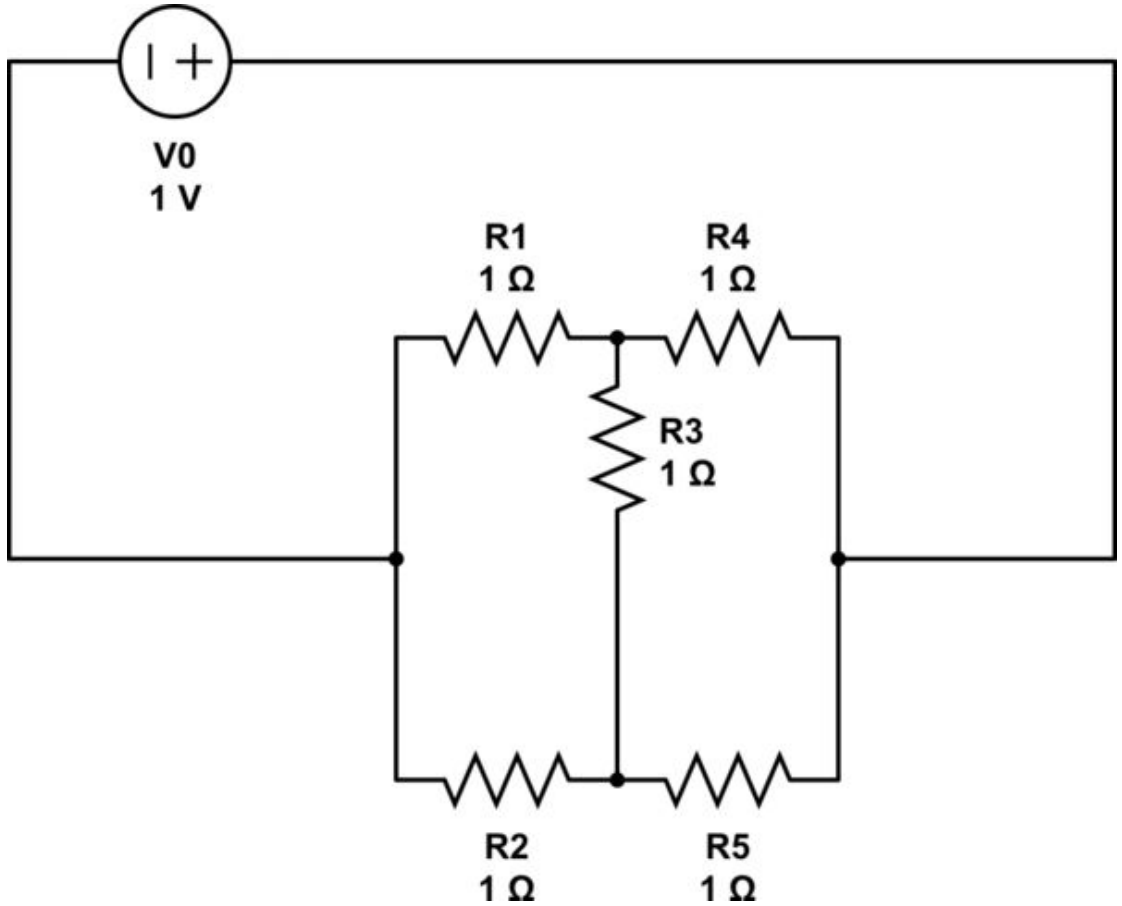
Correct Answer:

B

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Calculating Resistance



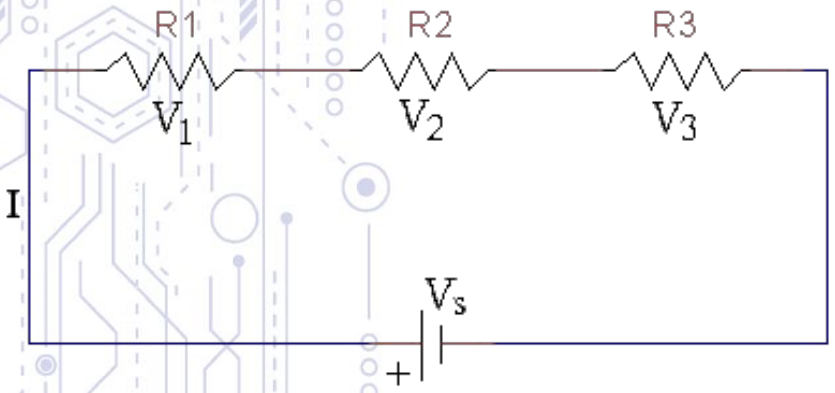
How do we determine total resistance (R) when provided a circuit diagram?

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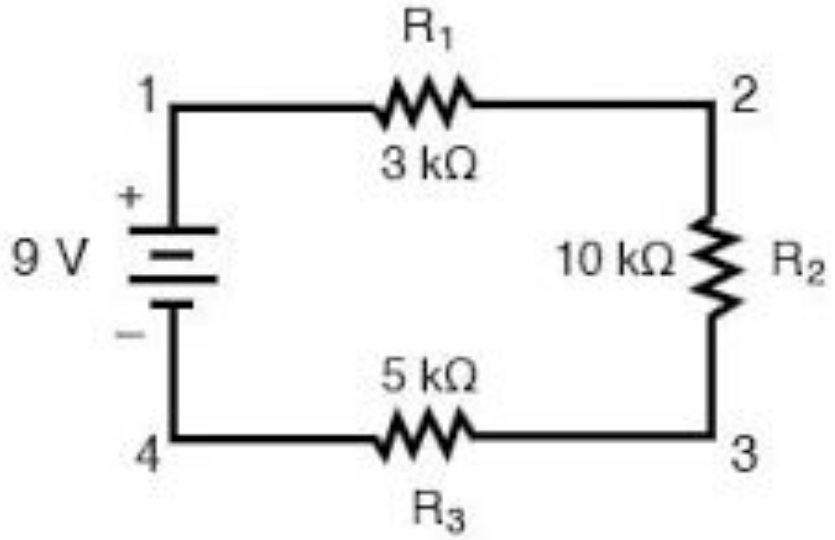
Resistors in Series



- If the resistors are in series (shown left), just sum all of the values together!
- $R_T = R_1 + R_2 + R_3 \ldots$

Check Your Understanding

Question 1/1

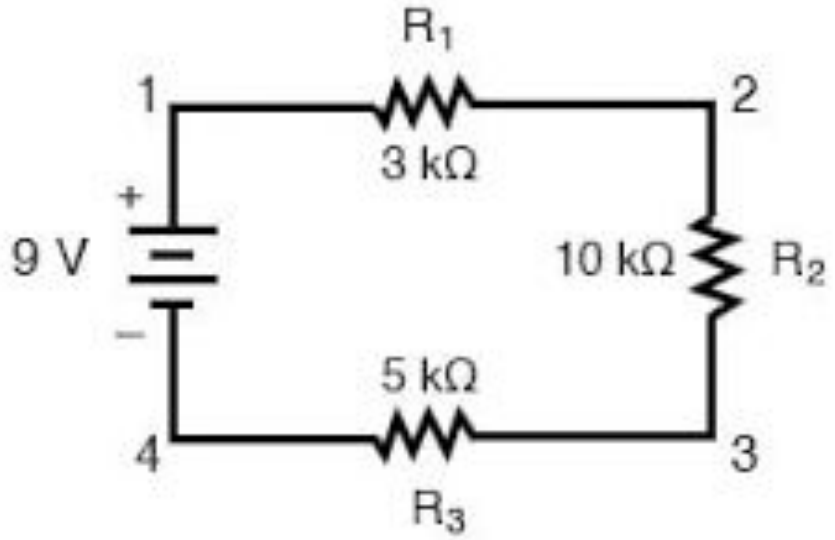


Calculate the total resistance of this circuit.

Note: Disregard the numbered labels in the corners of the diagram for now, but take note of them for later.

Check Your Understanding

Question 1/1

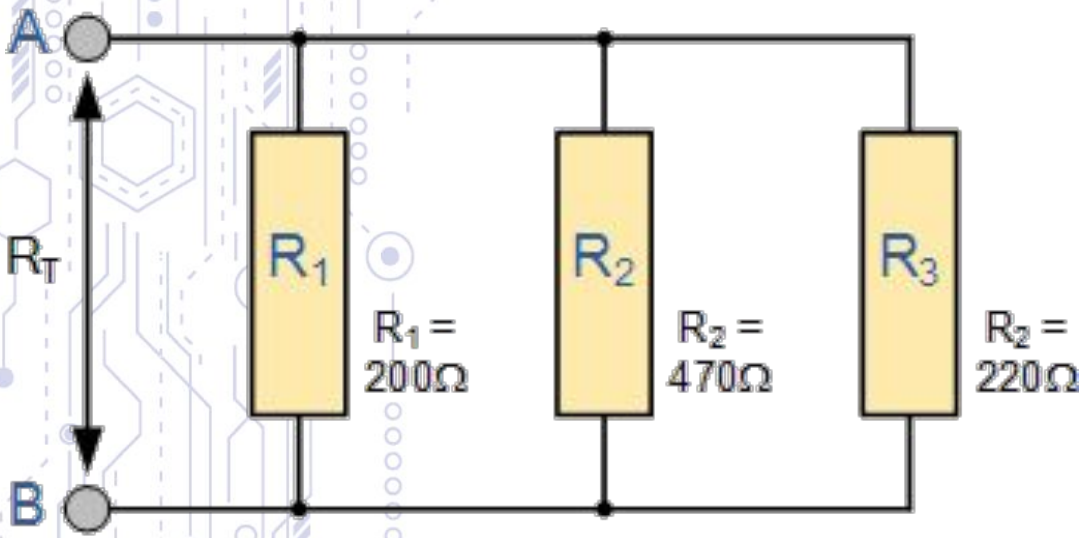


All of the resistors are in series, so we directly add them together.

$R_T = 3k\Omega + 10k\Omega + 5k\Omega$

$R_T = 18k\Omega$

Resistors in Parallel



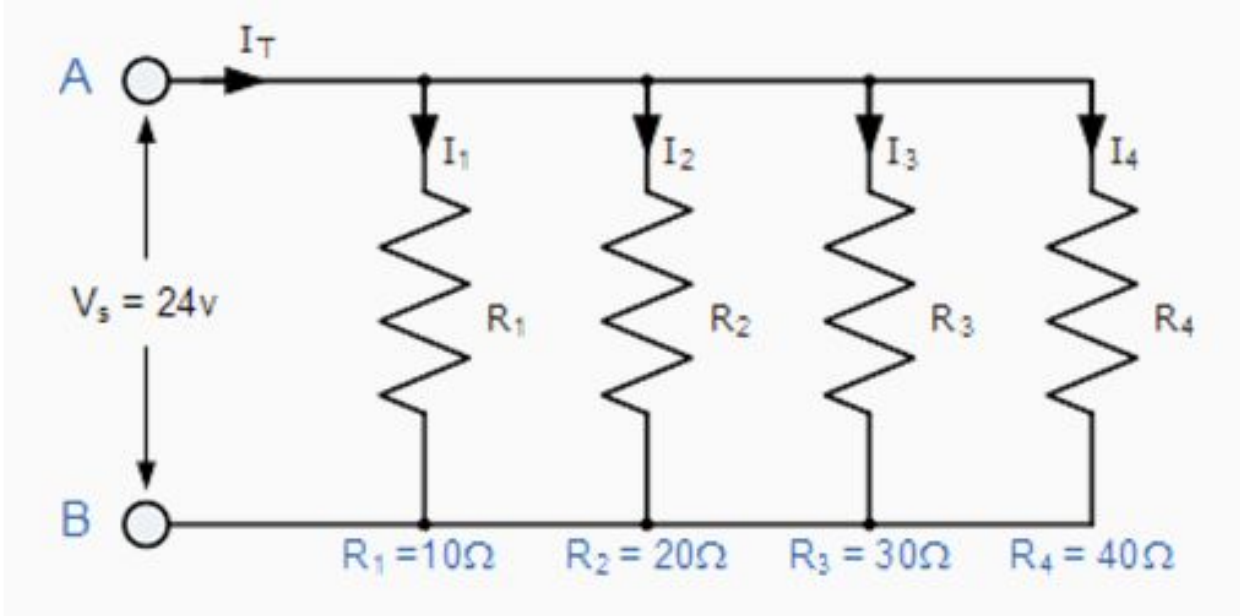
- If the resistors are in parallel, the formula is a bit different
- $1/R_T = 1/R_1 + 1/R_2 + 1/R_3 \ldots$

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Check Your Understanding

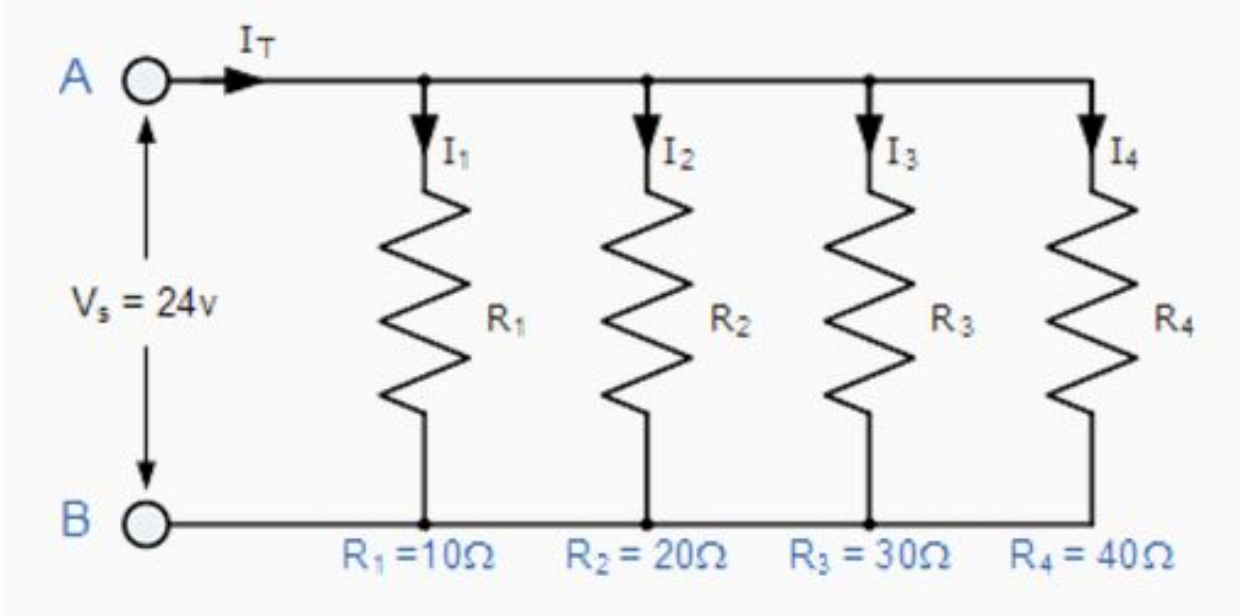
Question 1/1



Calculate the total resistance of this circuit.

Check Your Understanding

Question 1/1



All of the resistors are in parallel, so we use the formula:

$1/R_T = 1/10\Omega + 1/20\Omega + 1/30\Omega + 1/40\Omega$

$1/R_T = 5/24\Omega$

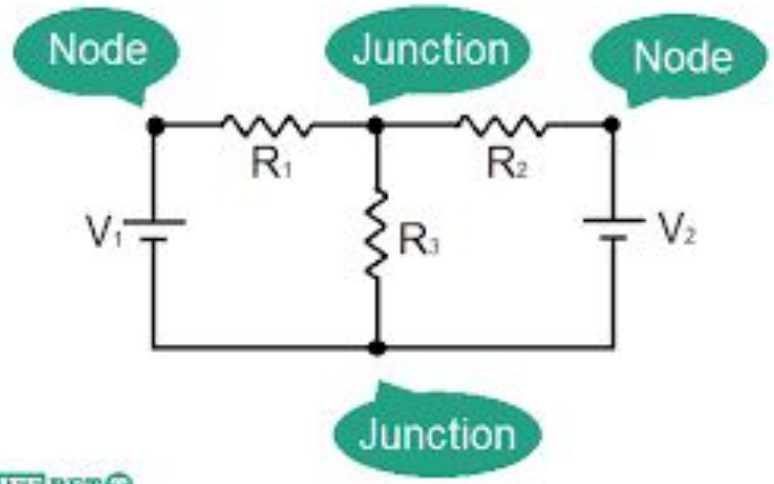
$R_T = 24/5\Omega = 4.8\Omega$

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Nodes

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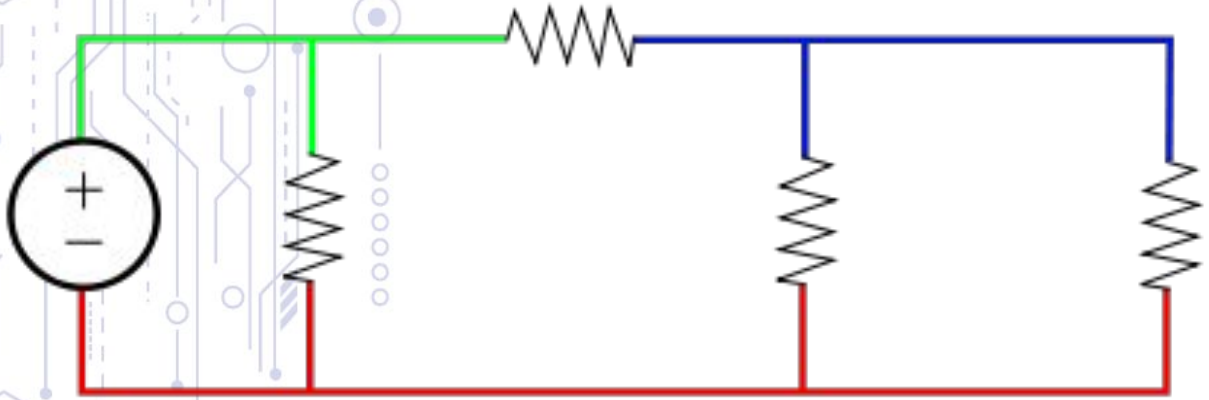
Nodes



- We can break down a circuit into an expression of connected nodes
- This is helpful for circuit debugging and design purposes

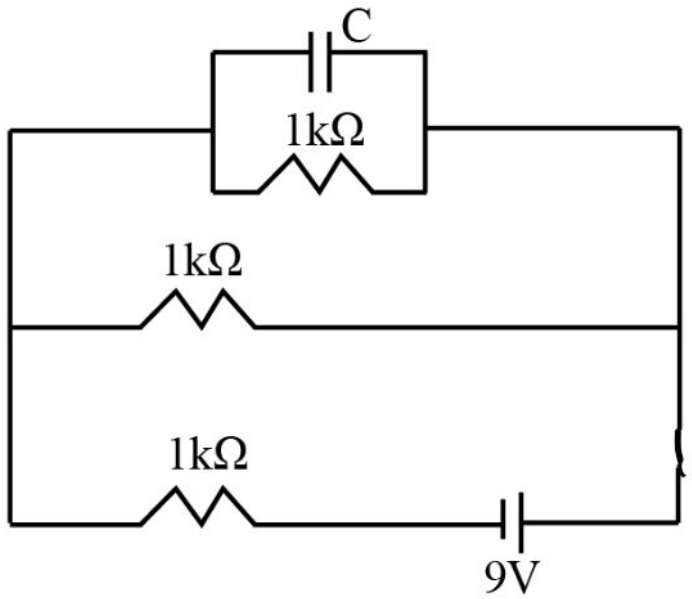
Nodes

- We define a node as the idealized wire (0 ohms of resistance) between circuit elements
- In other words, places where the voltage is the same



Check Your Understanding

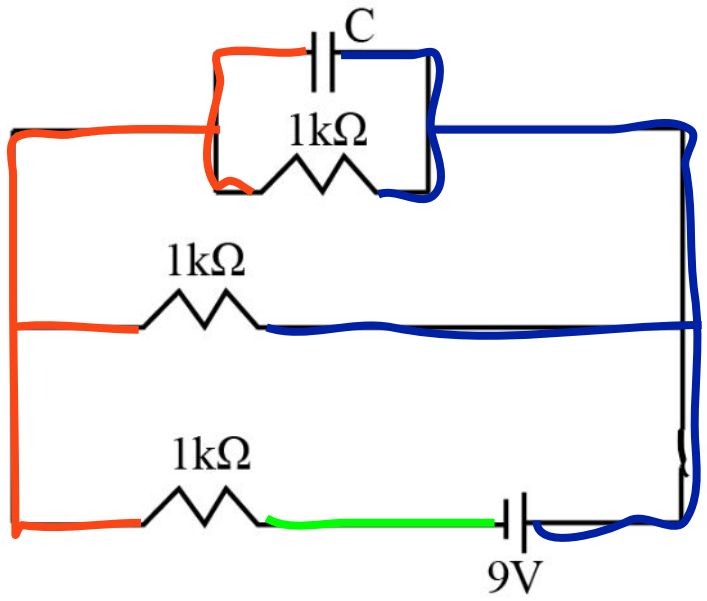
Question 1/3



Identify how many nodes are in the circuit diagram.

Check Your Understanding

Question 1/3

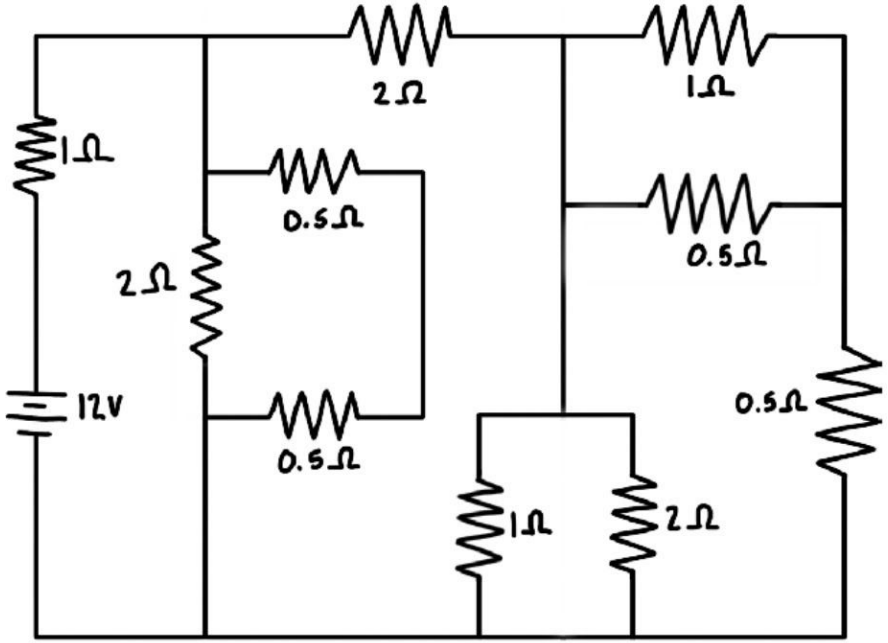


There are 3 nodes!

Remember that each node contains the wires *between* circuit elements, not the elements themselves.

Check Your Understanding

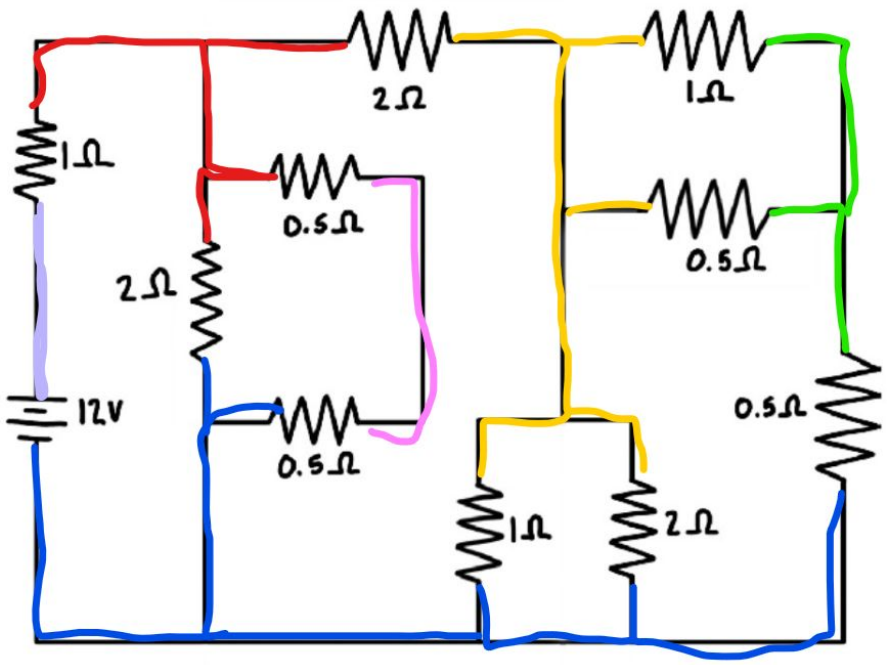
Question 2/3



Identify how many nodes are in the circuit diagram.

Check Your Understanding

Question 2/3

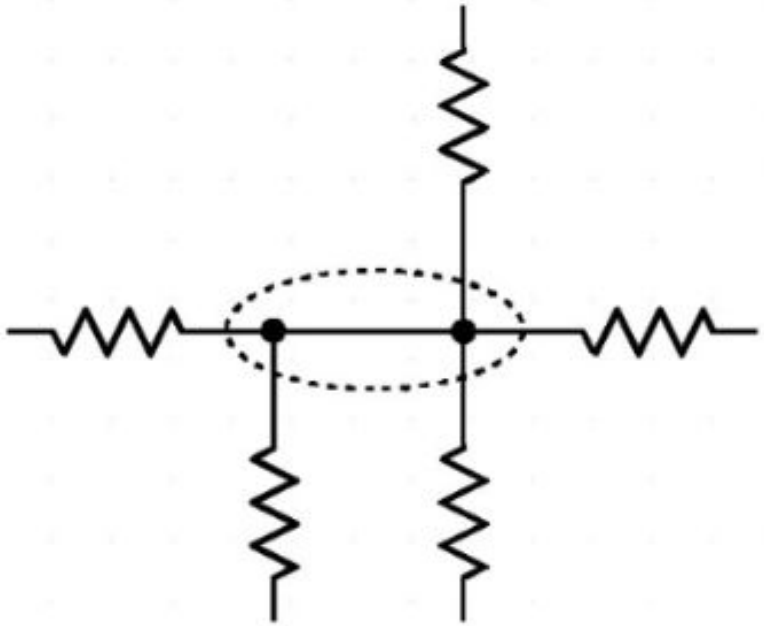


There are 6 nodes!

Resistors, capacitors, LEDs, and batteries are all examples of circuit elements.

Check Your Understanding

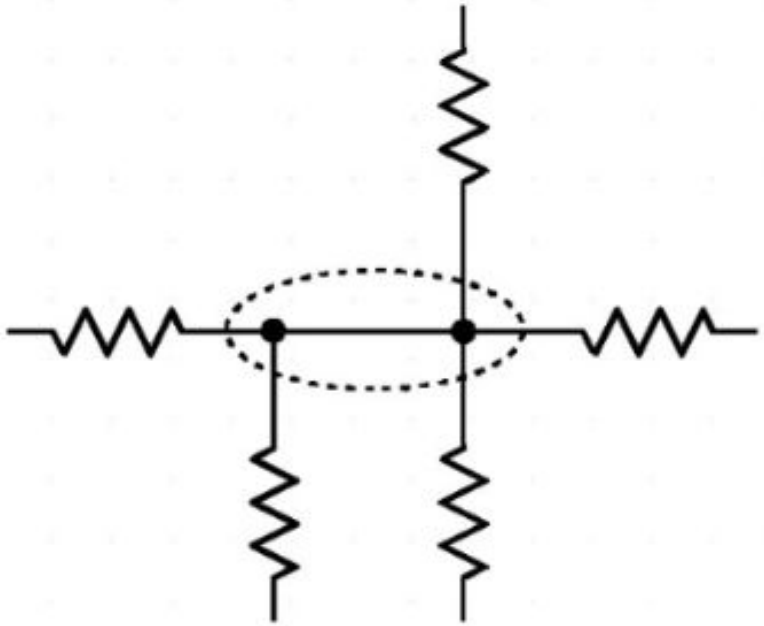
Question 3/3



What is the potential voltage difference between these two points on the circuit diagram?

Check Your Understanding

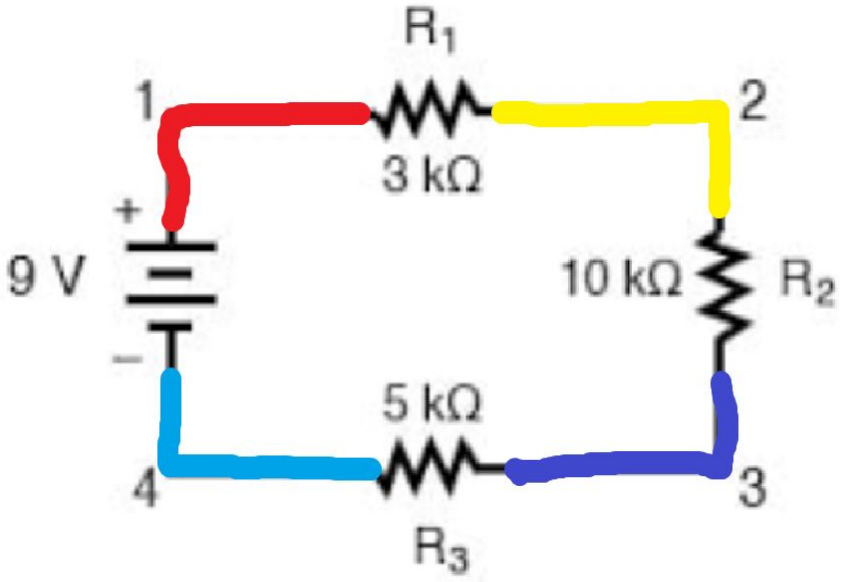
Question 3/3



The difference is 0 Volts because the two points are on the same node!

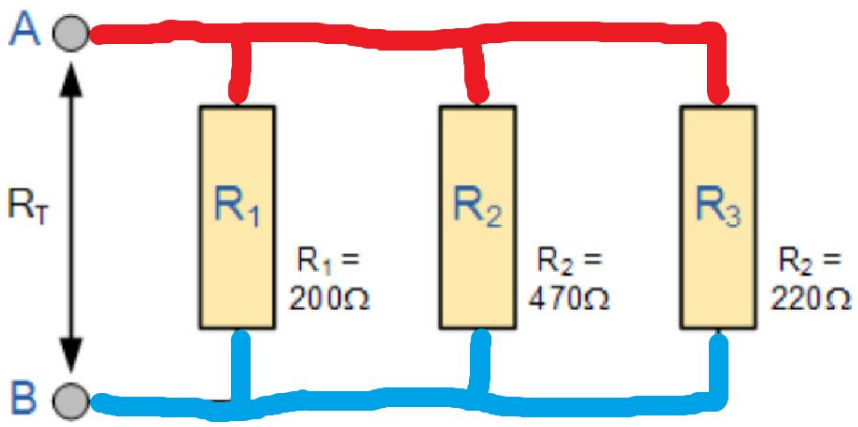
In this example, the two points on the schematic are said to be continuous

Nodes in Series Circuits



- A resistor in series will be connected to two different nodes
- Example: R_1 is connected to nodes 1 and 2, R_2 is connected to nodes 2 and 3, and so on

Nodes in Parallel Circuits

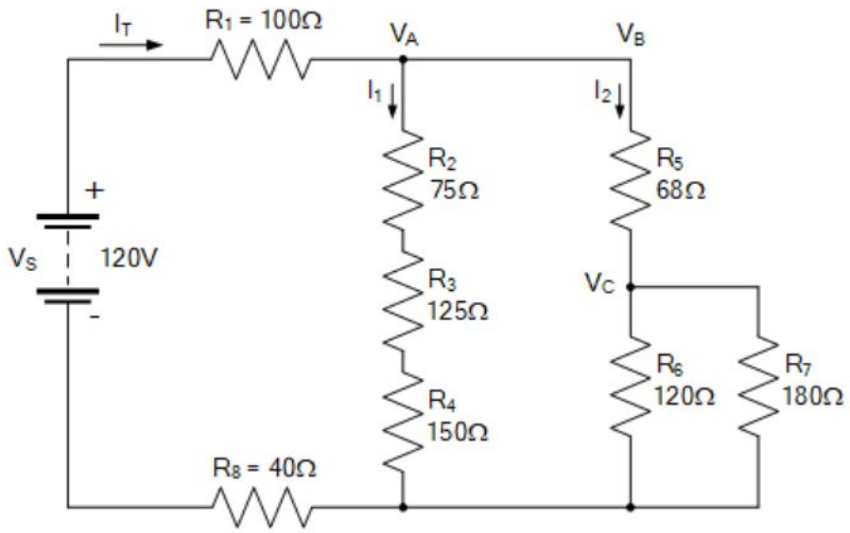


- Resistors in parallel will be connected to the same nodes
- Example: R_1, R_2, and R_3 are all connected to node 1 and node 2

Putting it All Together

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Identify the total resistance in the circuit diagram.



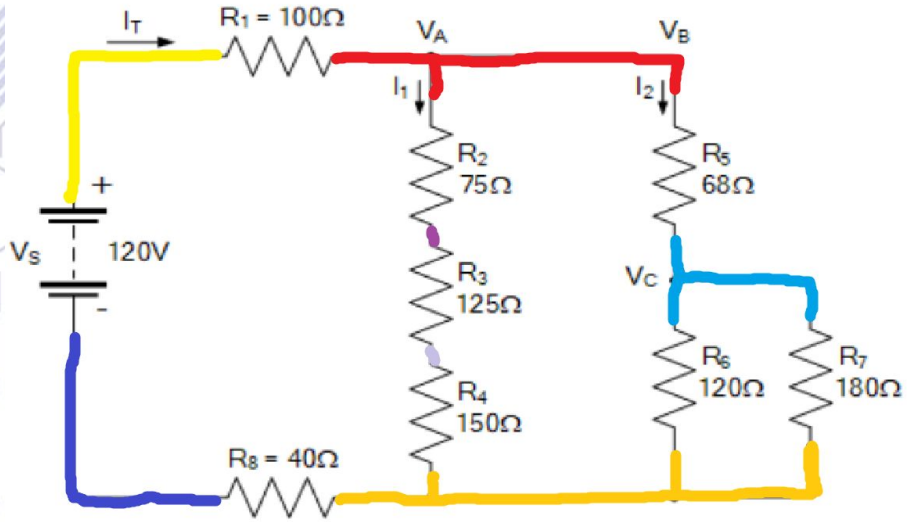
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Putting it All Together

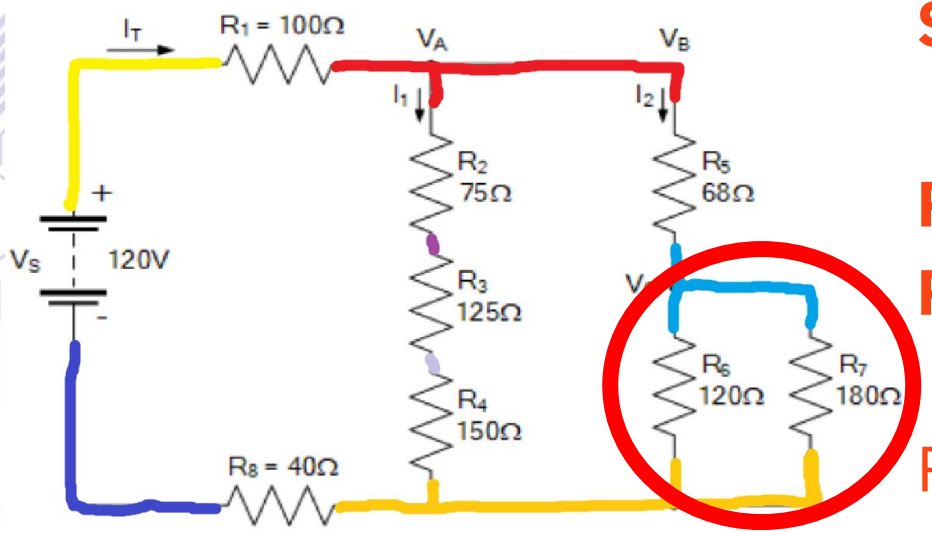
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Step 1: Count the nodes



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Putting it All Together



Step 1: Count the nodes

Step 2: Start simplifying

R6 and R7 share the same nodes!

Resistors are in <u>parallel</u>

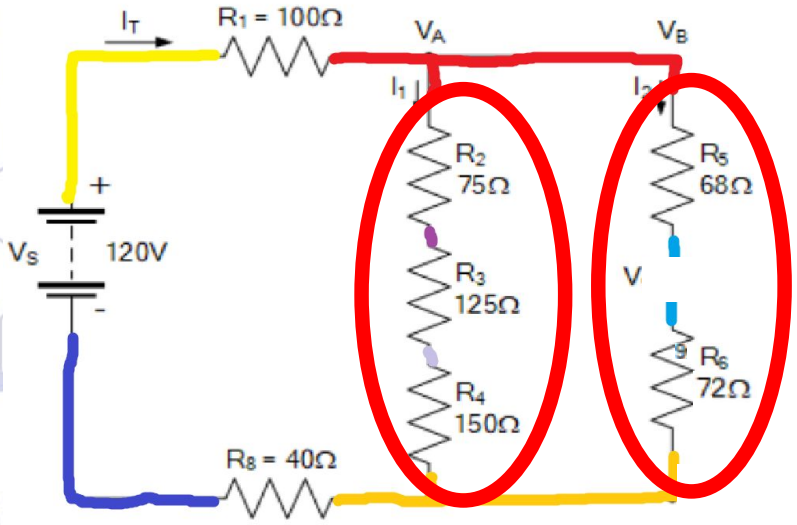
$R_9 = 1 / (1/R_6 + 1/R_7)$

$= 1 / (1/120\Omega + 1/180\Omega)$

$= 72\Omega$

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Putting it All Together



Step 1: Count the nodes

Step 2: Start simplifying

Step 3-4: Continue simplifying

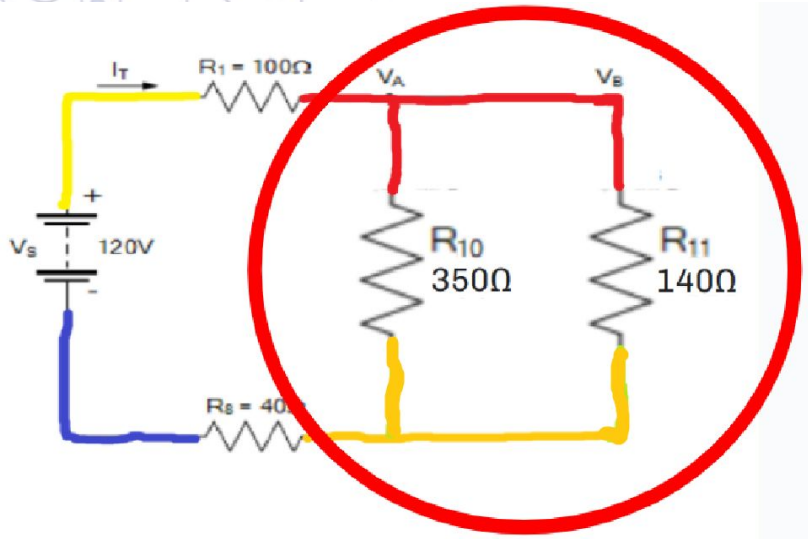
Can combine the resistors that share nodes without branches

$$R_{10} = R_2 + R_3 + R_4$$
$$= 75\Omega + 125\Omega + 150\Omega$$
$$= 350\Omega$$

$$R_{11} = R_5 + R_9$$
$$= 68\Omega + 72\Omega$$
$$= 140\Omega$$

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Putting it All Together



Step 1: Count the nodes

Step 2: Start simplifying

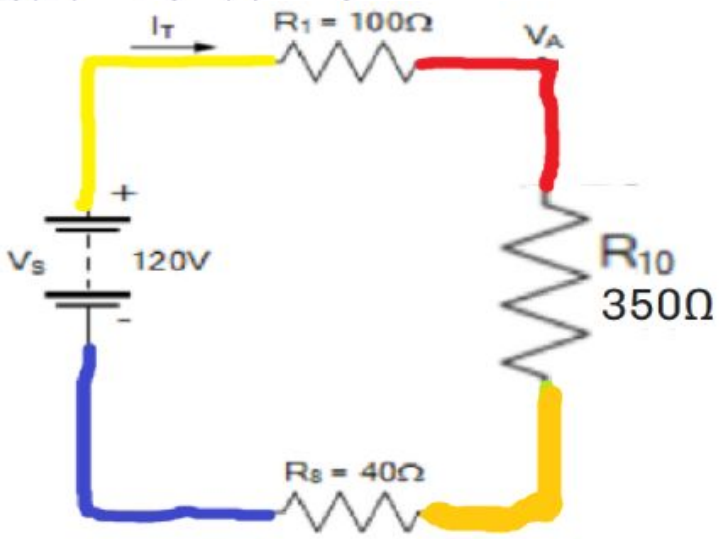
Step 3-4: Continue simplifying

R_{10} and R_{11} share the same nodes!
Resistors are in <u>parallel</u>

$R_{12} = 1 / (1/R_{10} + 1/R_{11})$
$= 1 / (1/350\Omega + 1/140\Omega)$
$= 100\Omega$

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Putting it All Together



Step 1: Count the nodes

Step 2: Start simplifying

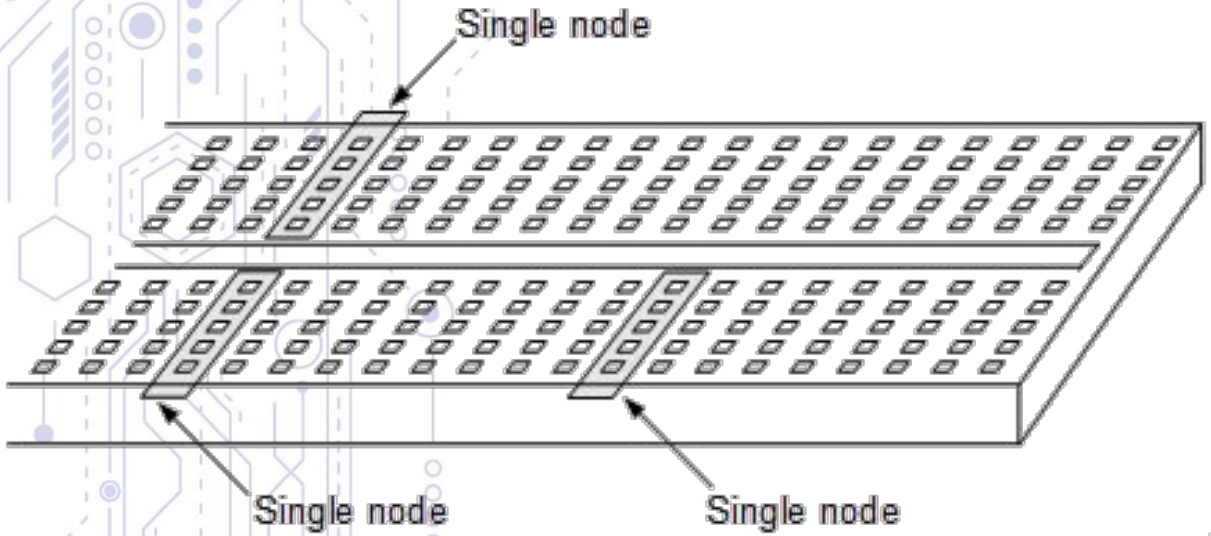
Step 3-4: Continue simplifying

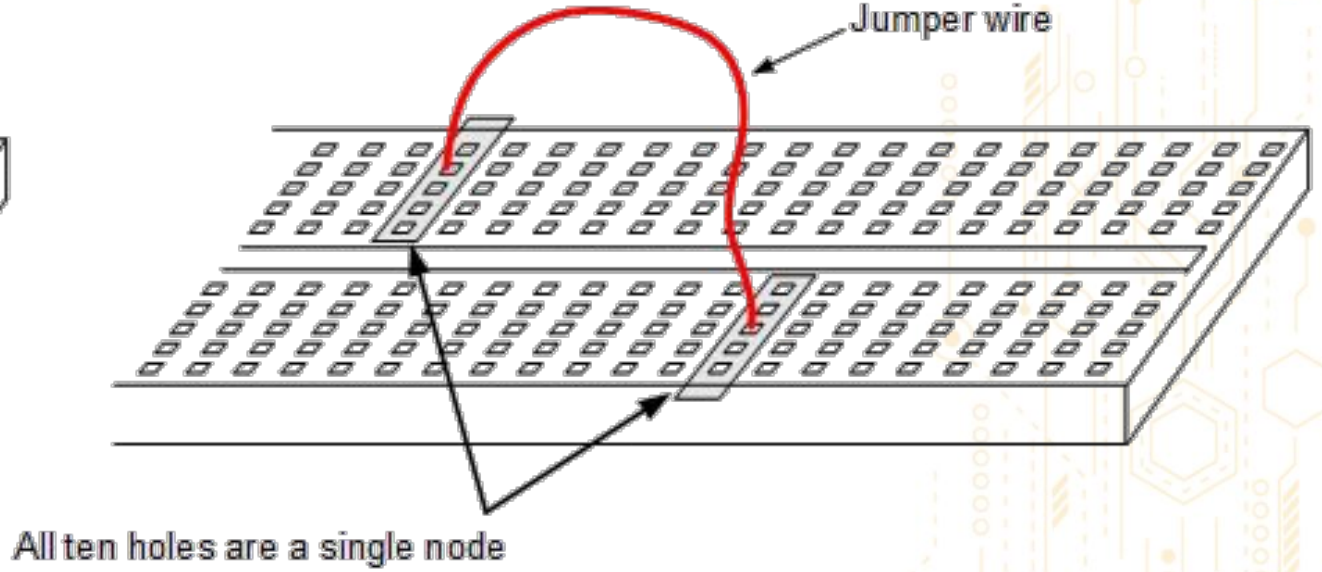
Step 5: Treat the circuit like a normal series circuit

$$R_T = R_1 + R_{12} + R_8$$
$$= 100\Omega + 100\Omega + 40\Omega$$
$$= \mathbf{240\Omega}$$

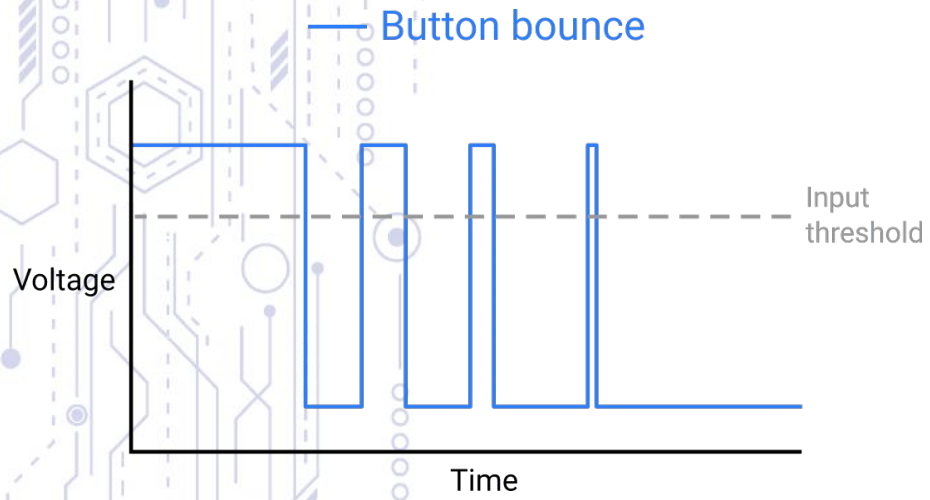
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Nodes: An OPS Example



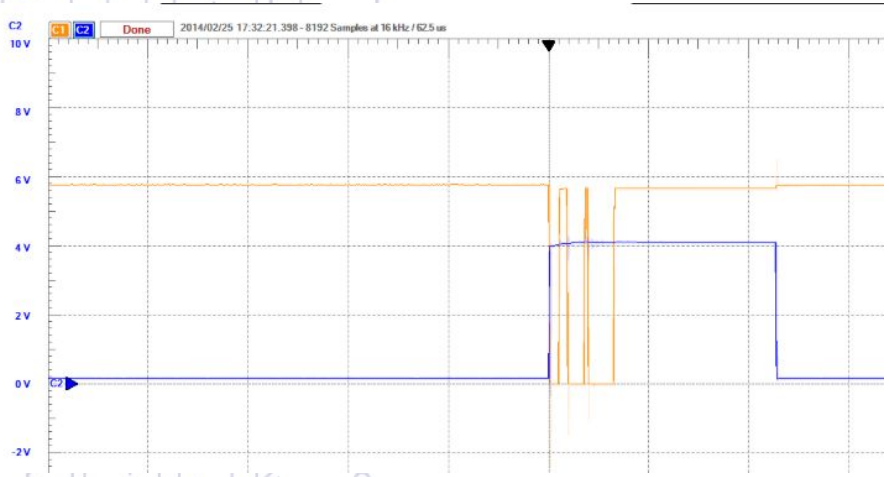


Bouncing



- Real life switches and buttons are not perfect
- When you press a switch or button, the signal bounces before settling

Bouncing

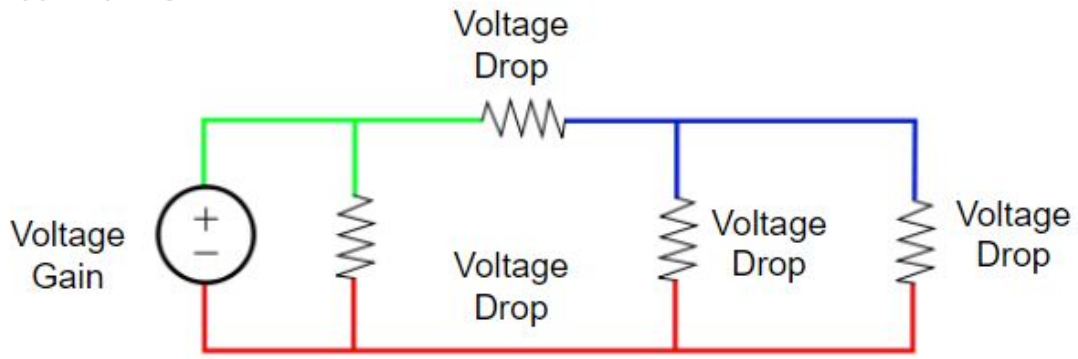


- This unstable signal can cause unintended behavior, flipping between ‘on’ and ‘off’ several times

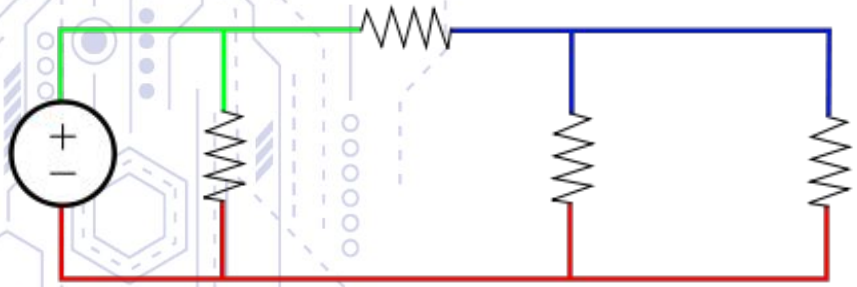
Circuit Debugging Methodologies

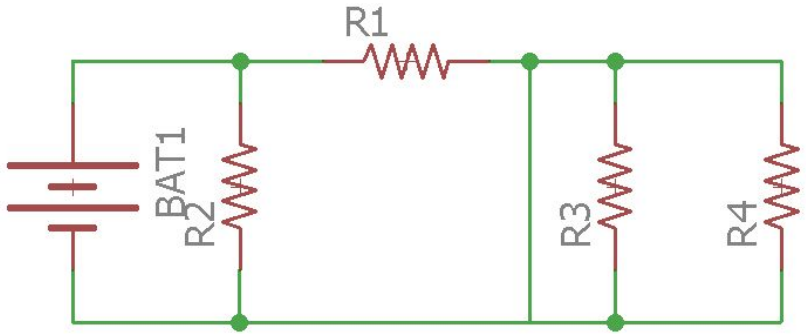
Voltage Drop & Gain

- When you look at the voltage before and after a circuit element, there is often a voltage drop or gain



Voltage Drop & Gain





- Typically, nodes have different voltages
- So, if two different nodes have the same voltage, it's usually indicative of a problem

Multimeters

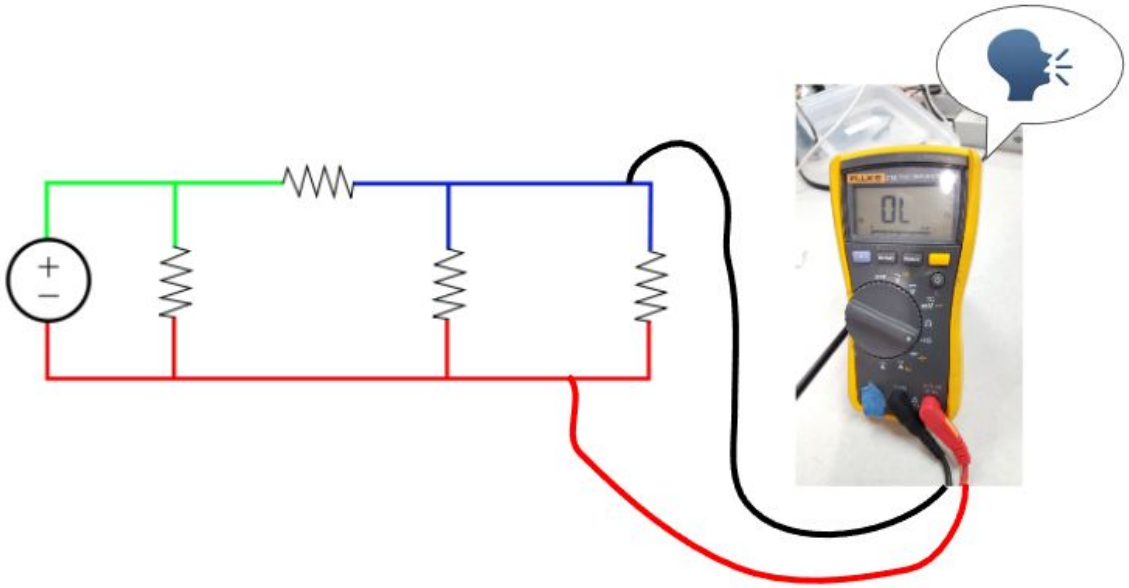


- Multimeters can typically measure voltage, resistance, and current
- This means that they can be used as a voltmeter, ohmmeter, and ammeter

How to Use a Multimeter

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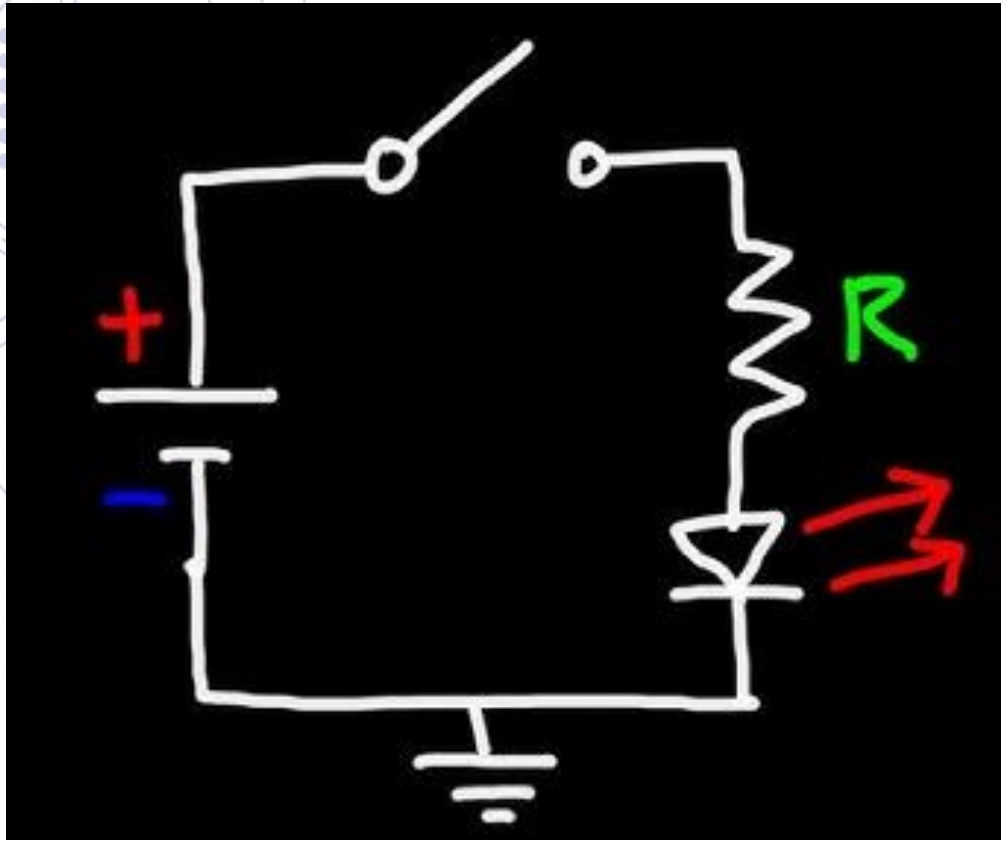
- We want to test for a voltage difference across elements

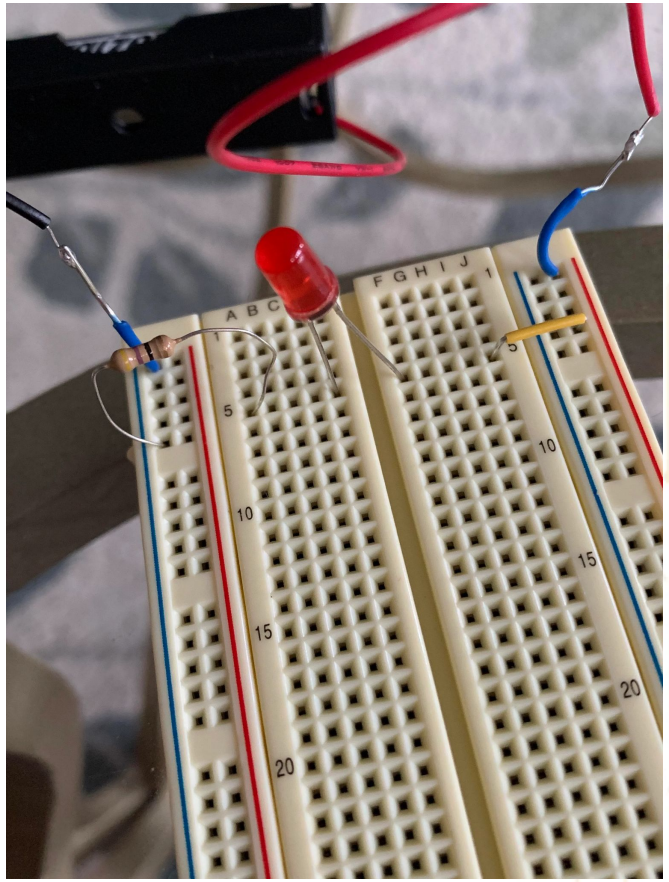


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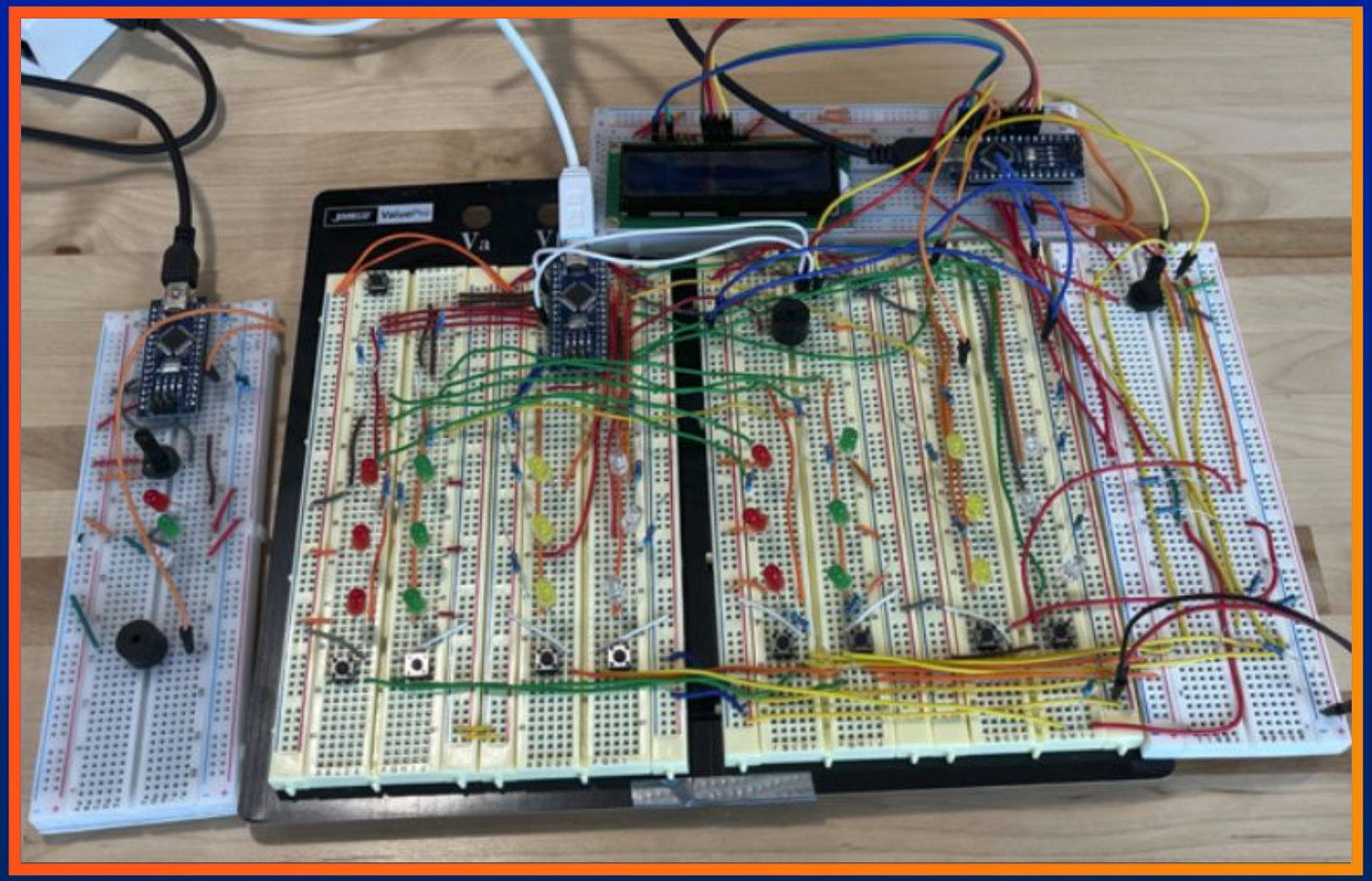
Simple Debugging





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Real Debugging

2024 OPS Capstone Project

Carter Amaba (IEEE-UF President), **Landon Amaba, Nick Lucindo** (Former OPS Coordinators)

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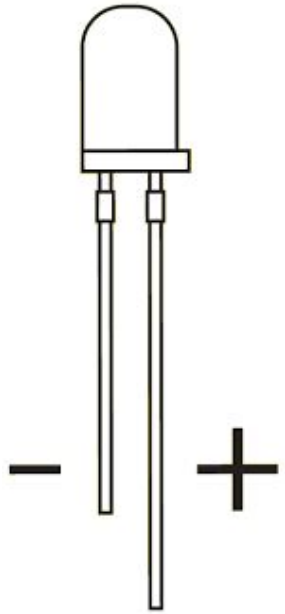
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Project 1: LED Circuits

Project 1: LED Circuits

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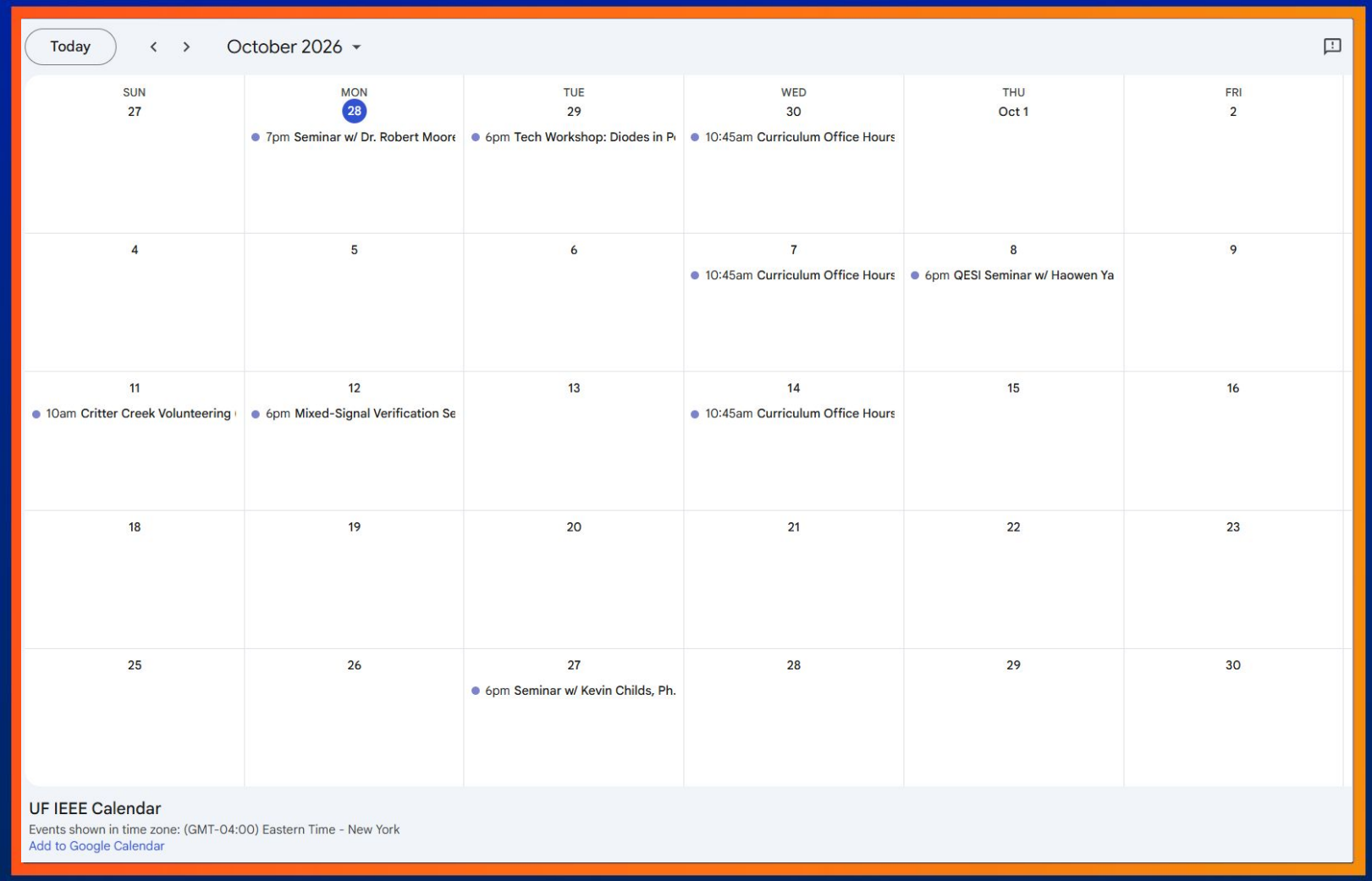


- Introduction to OPS kits
- Understand how to translate a circuit diagram into a breadboard prototype
- Understand how to design a circuit diagram given a list of specifications
- Understand how the potential difference between the nodes of an LED causes it to turn on
- Understand common LED circuits and button configurations

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Keep up with IEEE!

Upcoming events:

9/30, 10/7: Curriculum OH

10/8: QESI Seminar

10/11: Critter Creek Volunteering

Reminders:

Project 1 due Oct. 13!

Sign up for L2: Soldering!

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